

具身智能

第1讲：导论

Junwei Liang
AI Thrust
HKUST (Guangzhou)

Embodied AI

Lecture 1: Introduction

Junwei Liang
AI Thrust
HKUST (Guangzhou)

课程安排： 第1 部分

- 课程网站

- <https://hkust-aiaa4220.github.io/fall2026/lecs.html>
- 可查看课程日程、作业链接、课件下载、课程政策等
- 课件会在每次上课前发布

你现在就可以从日程网站下载今天的课件， 方便做笔记

Embodied AI

AIAA 4220, Fall 2026

Instructors: [Junwei Liang](#)



香港科技大学 (广州)
THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

Home	Schedule
Lecture:	Date and Time: 09:00AM - 11:50AM, Monday Location: Rm 233, W1 Websites: Instructure Zoom: 628-334-1826 PW: 12345 (In-person expected) (Recordings will be available after each class)

Course Description: This course aims to provide students with key principles and algorithms to build modern Embodied AI systems. Key topics include machine perception, embodied navigation and manipulation algorithms. Through this course, students will learn and practice the foundational principles, techniques, and tools to build new Embodied AI systems.

Target Audience/Prerequisites: This is a course primarily for junior and senior students.

Course Work:

Grading	Grading will be based on best 5 of 6 in-class quizzes (25%), one paper presentation (25%), one machine perception homework (25%) and one simulation project (25%).
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In-Class Quizzes

- The in-class quizzes are designed to complement the lectures and making sure you understand the key knowledge about the course.
- There are 6 in-class quizzes over the semester. We will take your best 5 out of the 6 quizzes as your grade.

我们今天稍后会讲课程政策

Logistics: Part 1

- Course website

- <https://hkust-aiaa4220.github.io/fall2026/lecs.html>
- You can see the **course schedule, homework links, download slides, policy**, etc.
- The slides will be available before each class

You can download the slides now from the schedule website for today’s lecture if you want to take note on it

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We will go over the course policy later today

本学期授课教师与助教 (AIAA 4220)



Junwei Liang
junweiliang@hkust-gz.edu.cn
主讲教师

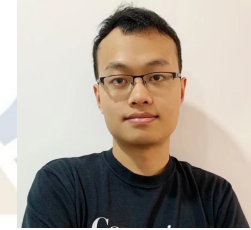
- 上课时间: 每周一 9:00 - 11:50
- 每小时会休息一次

助教:

Yuxin He
yhe469@connect.hkust-gz.edu.cn

Jinghao Huang
jhuang401@connect.hkust-gz.edu.cn

Your Instructor and TAs This Semester (AIAA 4220)



Junwei Liang
junweiliang@hkust-gz.edu.cn
Course lecturer

- Our lectures are scheduled on Monday every week from 9:00 am. to 11:50 am.
- We will take a break every hour

Your TA:

Yuxin He
yhe469@connect.hkust-gz.edu.cn

Jinghao Huang
jhuang401@connect.hkust-gz.edu.cn

提纲

- 先介绍一下我
- 什么是具身智能?
- 机器人简史
- 为什么现在学具身智能?
- 课程内容概览
- 课程安排

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- Logistics

先介绍一下我 ...

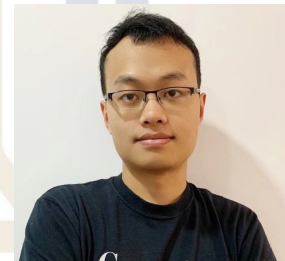


Junwei Liang
junweiliang@hkust-gz.edu.cn
AIAA 4220 主讲教师

- 我来自广州
- 先后在北京（人大）、匹兹堡（CMU）学习，并在 Mountain View（Google）实习
- 后来去了上海（腾讯）
- 现在回来了！



A bit about me ...



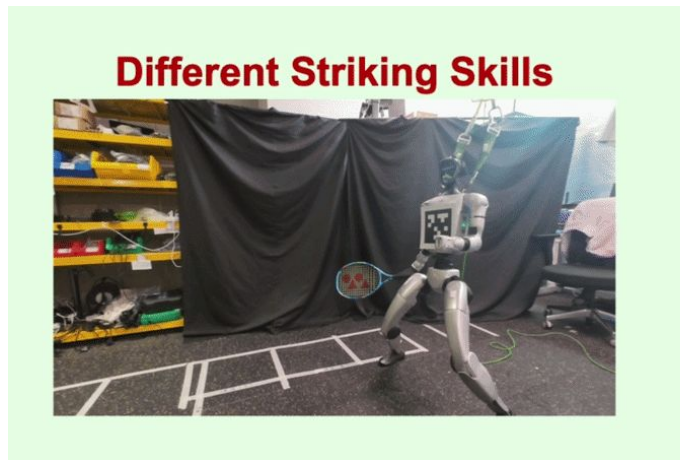
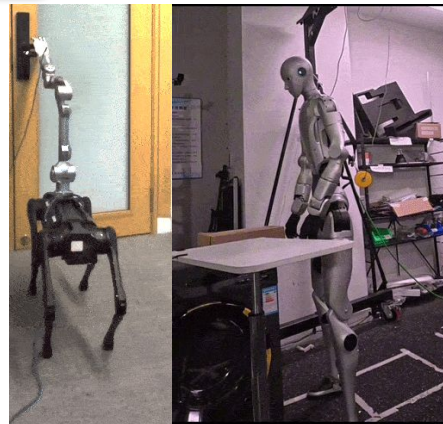
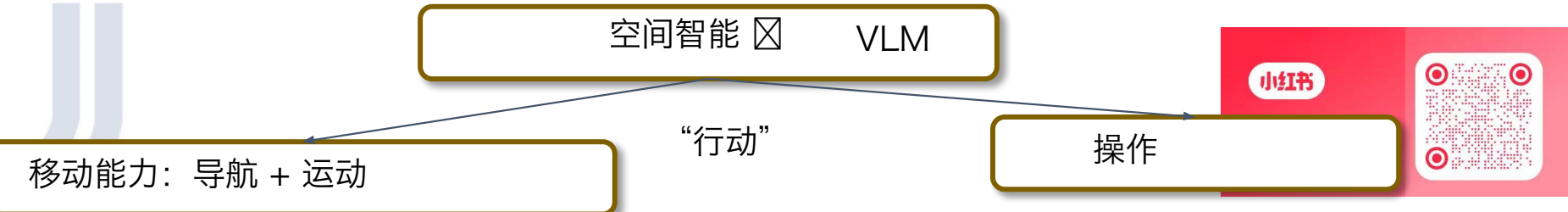
Junwei Liang
junweiliang@hkust-gz.edu.cn
AIAA 4220 course lecturer

- I'm from Guangzhou
- I studied in Beijing (@RUC), Pittsburgh (@CMU) and interned in Mountain View (@Google)
- I moved to Shanghai (@Tencent)
- And now I'm back!



我们实验室是大湾区领先的具身智能实验室

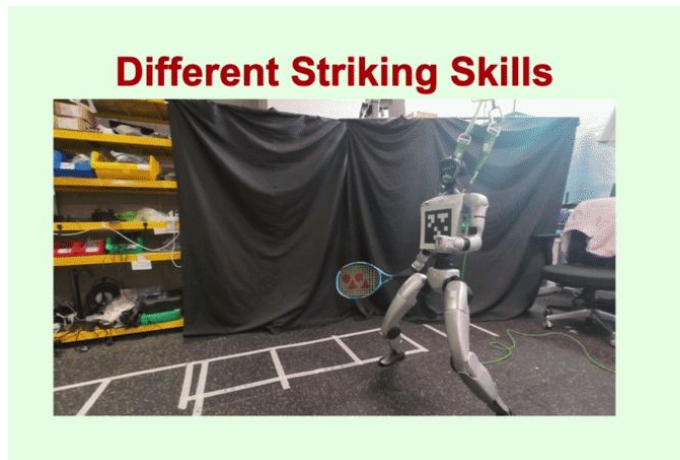
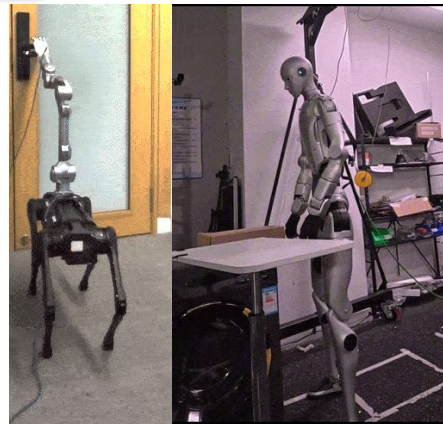
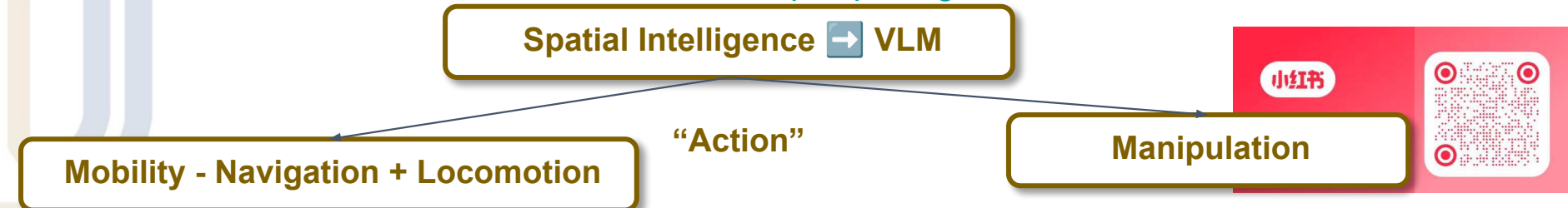
• 更多演示见实验室网站: <https://precognition.team/> 也可看我们的小红书!



今天还会介绍实验室更多前沿研究

Our Lab is the leading Embodied AI lab in the greater bay area

• Check our lab website for more cool demos: <https://precognition.team/> Or check out our 小红书!



Agile Locomotion

Manipulation with dex hand

Athletic Intelligence

More cutting-edge research of our lab will be shown today

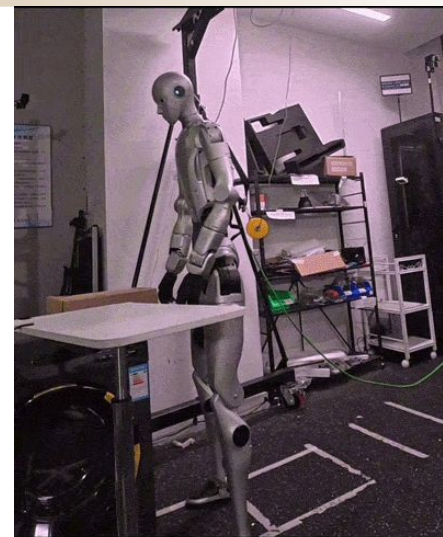
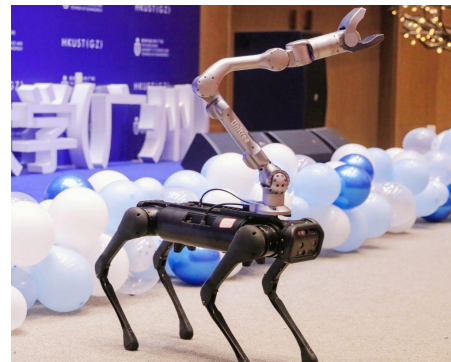
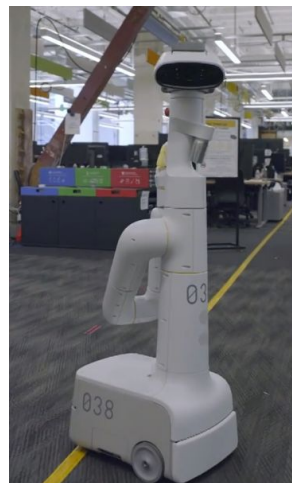
提纲

- 先介绍一下我
- 什么是具身智能?
- 机器人简史
- 为什么现在学具身智能?
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- **What is Embodied AI?**
- History of Robotics
- Why study Embodied AI now?
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- Logistics

什么是具身智能？



机器人？双足、轮式、四足……

看看它们怎么动

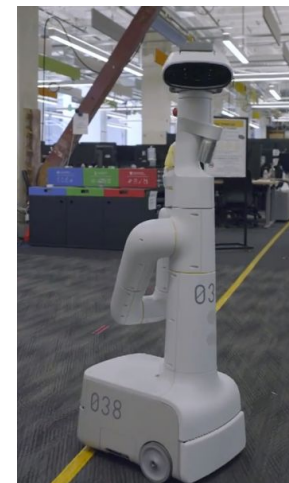
* <https://www.youtube.com/watch?v=Gm6dZ1q06ks>

* https://www.youtube.com/watch?v=-e1_QhJ1EhQ

What is Embodied AI?



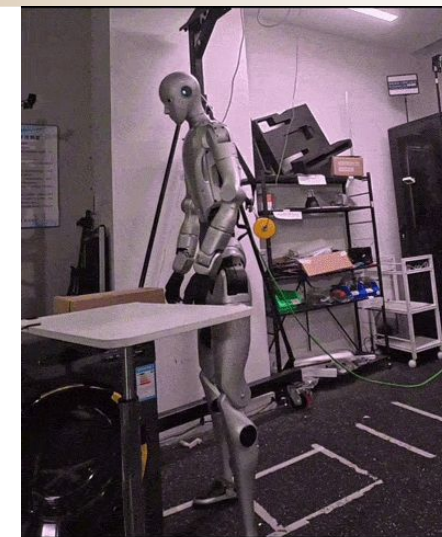
Boston Dynamics



Single-arm Robot



Unitree Robotics



Humanoid

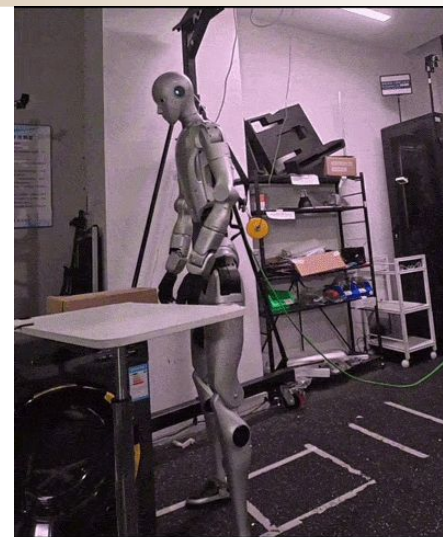
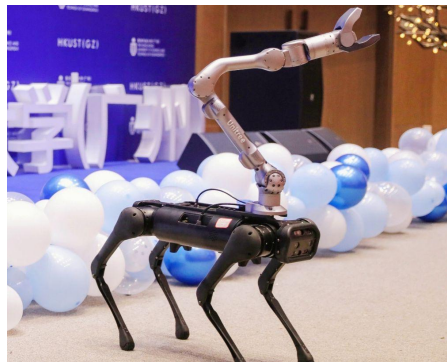
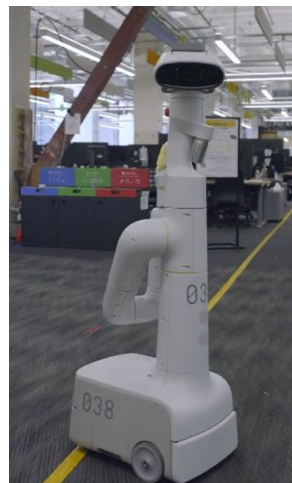
Robots? Two-legged, wheeled, four-legged...

See how they move

* <https://www.youtube.com/watch?v=Gm6dZ1q06ks>

* https://www.youtube.com/watch?v=-e1_QhJ1EhQ

什么是具身智能？



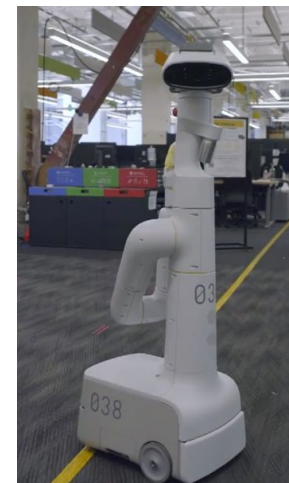
机器人？任何能自动完成任务的机器！

- 可以基于预设规则，也可以通过机器学习……

What is Embodied AI?



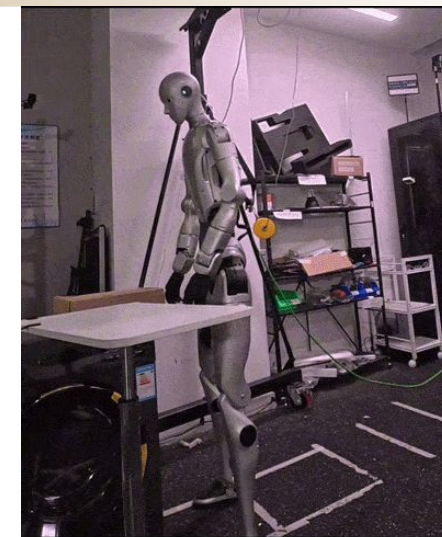
Boston Dynamics



Single-arm Robot



Unitree Robotics



Humanoid

Robots? Any machine that can perform a task automatically!

- Could be based on prespecified rules or through machine learning..

什么是具身智能？

- 具身智能是能感知环境、并在真实世界中据此完成任务的机器人。
- 关键智能属性
 - 适应性（应对世界中的噪声）
 - 从数据中学习

洗衣机算具身智能吗？不算。
它没有环境感知

清洁机器人算具身智能吗？也许算。它能检测障碍
并打扫房间



What is Embodied AI?

- Embodied AI is a robot capable of **sensing its environment** and **performing its tasks accordingly** in the real world.
- Key intelligent properties
 - Adaptability (account for noise in the world)
 - Learning from data

Are washing machines considered Embodied AI? No.
There is no environment sensing

Are clearing bots Embodied AI? Maybe. It detects
obstacles and clean the house



什么是具身智能？

- 具身智能是机器学习与机器人学的交叉
 - 中文：具身智能
- 研究如何用感知数据训练最优方案，让机器人行动并达成目标
 - 感知与决策是关键
 - 可能的未来应用

○ 通用家用机器人！



What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
 - In Chinese: “具身智能”
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future application
 - General-purpose home robots!



具身智能 —— 为了什么？

- AI 应接管机器人研究者所说的三个 “D”

- 枯燥 (Dull)

- 重复性任务

- 肮脏 (Dirty)

- 危险 (Dangerous)

- 仓储机器人

- 清洁机器人

- 水下机器人

- 等等

Embodied AI - What for?

- The AI should take over the three “D”s (according to the robotics researcher)

- The dull

- Repetitive tasks

- The dirty

- The dangerous

- Warehouse robots

- Cleaning bots

- Underwater robots

- etc.

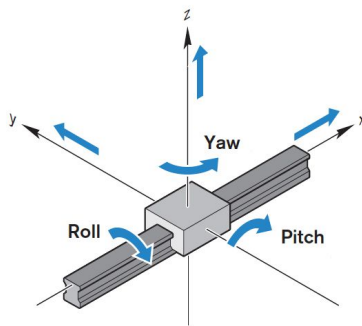
什么时候不需要机器学习

• 工业机器人

- 1970 年代开始普及
- 通常是单臂、不能移动
- 用最多 6 个自由度搬运物体

- Xyz: 位置
- 俯仰、偏航、滚转: 方向
- 被编程去做重复动作 (枯燥的事)

- 有时也很危险, 例如搬重物
- 重复、不变的流程



Define points P1–P5:

1. Safely above workpiece (defined as P1)
2. 10 cm Above bin A (defined as P2)
3. At position to take part from bin A (defined as P3)
4. 10 cm Above bin B (defined as P4)
5. At position to take part from bin B. (defined as P5)

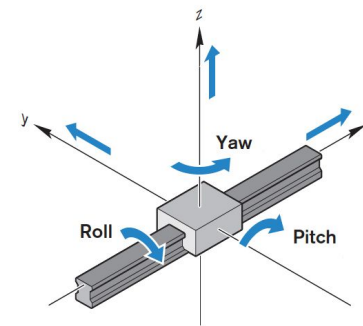
Define program:

1. Move to P1
2. Move to P2
3. Move to P3
4. Close gripper
5. Move to P2
6. Move to P4
7. Move to P5
8. Open gripper
9. Move to P4
10. Move to P1 and finish

When Machine Learning is **NOT** Needed

• Industrial Robot

- Populated from 1970s
- Usually a single arm with no mobility
- Move an object with up to 6 DoF
 - Xyz - position
 - pitch, yaw, roll - direction
- Programmed to do repetitive actions (The Dull)
 - Sometime dangerous, lifting heavy stuff
- **Repetitive, non-changing processes**



Define points P1–P5:

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7. Move to P5
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9. Move to P4
10. Move to P1 and finish

什么时候不需要机器学习（大多数时候……）

- 物理过程

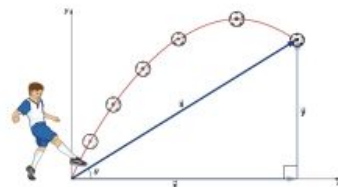
- 不同执行器的运动学与逆运动学
- 含义：机器人（部件）如何运动到目标位置
- 不必重新发明轮子。我们已有很好的物理过程模型

- 称为基于模型的控制

- 可以直接调用 API。当然仍可用机器学习去改进

- 例如强化学习！

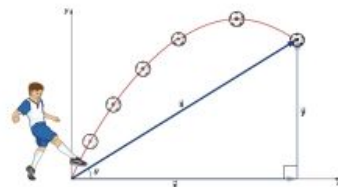
- 关键问题包括
Sim-to-Real，以及不完美执行的补偿



When Machine Learning is **NOT** Needed (for most of the time...)

- Physical processes

- **Kinematics and inverse kinematics** of different actuators
- Meaning: how robots (parts) move to a desired position
- We don't have to "reinvent the wheel". We have good models of physical processes
 - Called **Model-based Control**
- We can just call their APIs. Although you can still use machine learning to improve
 - With reinforcement learning!
 - Key problems include Sim-to-Real, accounting for the imperfect actuation



什么时候需要机器学习

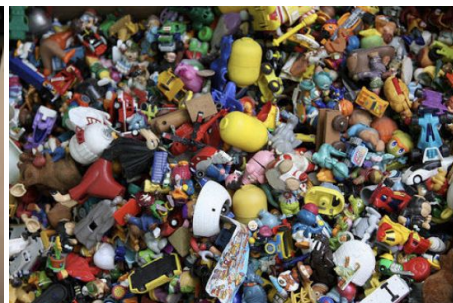
- 非结构化环境中的机器人



结构化/已知环境



环境



When Machine Learning **IS** Needed

- Robots in **unstructured environments**



Structured/known environment



Environment
uncertainty



Object variation

具身智能 vs. 自动化

Special-Purpose Robot Automation

没有机器学习时，通常叫自动化，而不是自主，并且需要人工标定



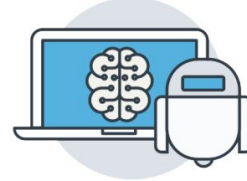
special-purpose behaviors

General-Purpose Robot Autonomy



general-purpose robots

Robot Learning



general-purpose behaviors

Embodied AI vs. Automation

Special-Purpose Robot Automation

Without machine learning, it is usually called automation, not autonomy



custom-built robots



human expert programming & manual calibration



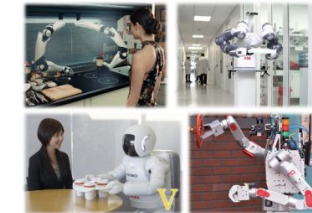
special-purpose behaviors

General-Purpose Robot Autonomy



general-purpose robots

Robot Learning



general-purpose behaviors

非结构化环境中的进展

- (新) 仓储机器人

- Boston Dynamics 2022 新产品
- 把箱子从卡车卸到仓库
- 适应多种箱子

■ 吸盘夹爪

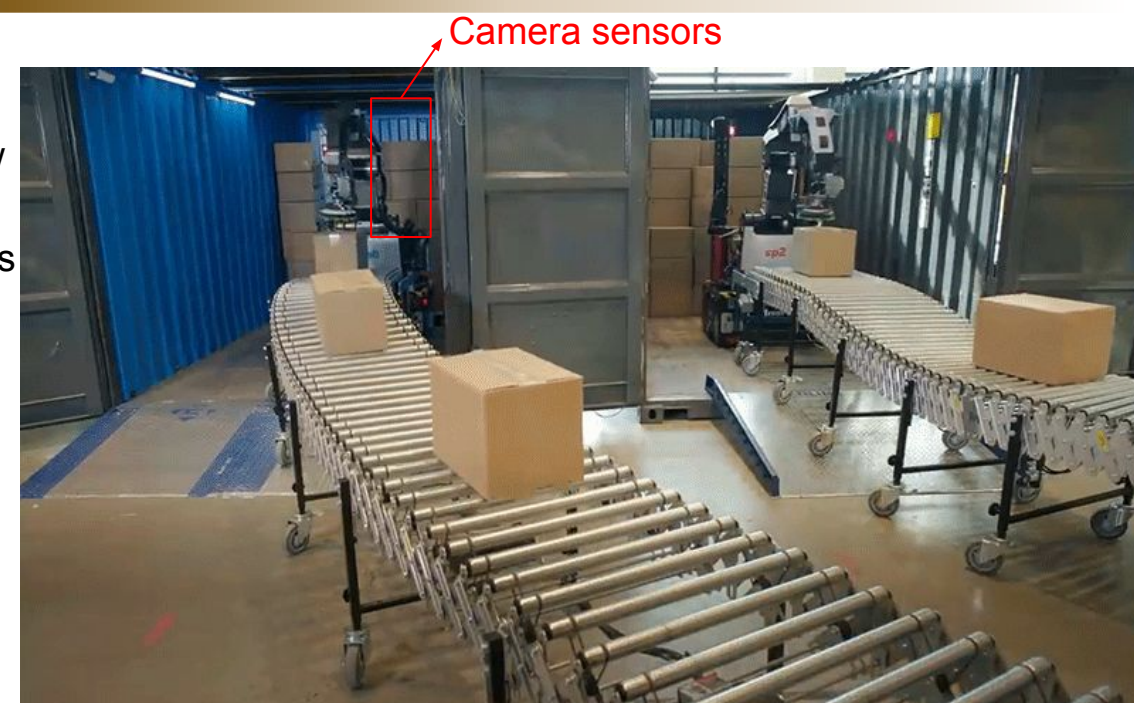


* 了解 Stretch: <https://www.youtube.com/watch?v=qgGH6Vk8nY8>

Advancement in Unstructured Environment

- (New) Warehouse Robot

- Boston Dynamics 2022 new product
- Unloading boxes from trucks to warehouses
- Adapt to many kinds of boxes
 - Suction gripper



* Learn more about Stretch: <https://www.youtube.com/watch?v=qgGH6Vk8nY8>

非结构化环境中的进展



- 我们已经在使用一些具身智能产品
 - 清洁机器人 — 关键技术: SLAM
 - 宠物猫砂盆 — 宠物识别 — 红外与重量传感器
 - 家用安防摄像头 — 行人检测与跟踪
 - 自动驾驶出租车
 - 配送机器人

Advancement in Unstructured Environment



- We are already using some form of Embodied AI products
 - Cleaning robots - key tech: SLAM
 - Pets litter box - pet recognition - infrared and weight sensor
 - Home Security camera - person detection and tracking
 - Self-driving taxi
 - Delivery robots

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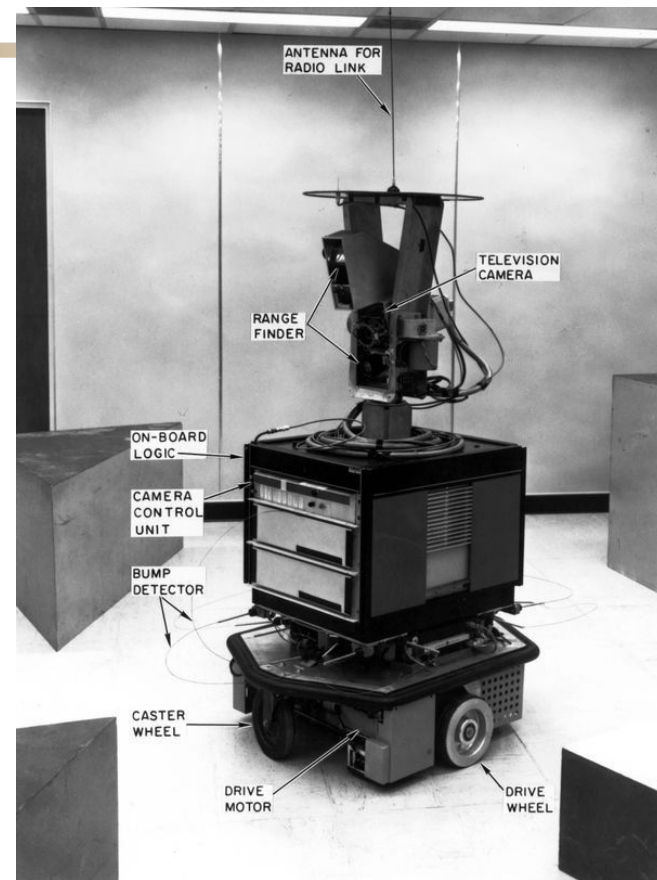
- A bit about me...
- What is Embodied AI?
- **History of Robotics**
- Why study Embodied AI now?
- Course content overview
- Logistics

机器人简史

- 第一台“真正的”机器人

- 1960 年代 — Shakey — A* 算法

- Charles Rosen, Stanford Research Institute

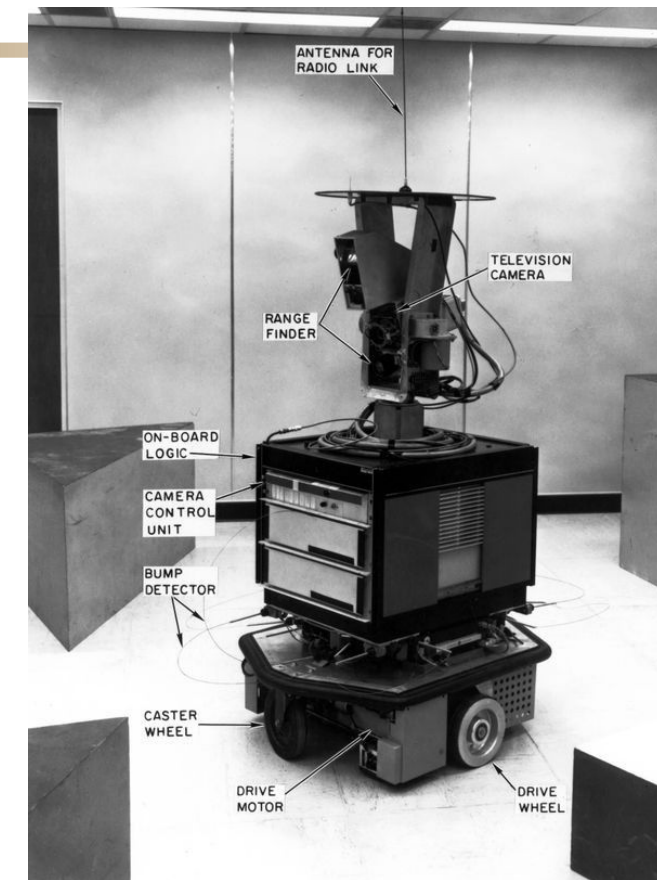


History of Robotics

- The first “real” robot

- 1960s - Shakey the robot - A* algorithm

- Charles Rosen, Stanford Research Institute



机器人简史

- 1980 年代 — 机械臂在自动化中普及



History of Robotics

- 1980s - Robotic Arms are populated in Automation



机器人简史

- 2000 年代

- 清洁机器人 — Roomba/iRobot — 2002

- 简单算法

- 简单碰撞传感器

- 不对房间建图

- 撞到物体或墙后随机改角度行走



长曝光照片显示 Roomba 清扫时走过的路径



History of Robotics

- 2000s

- Cleaning robots - Roomba/iRobot - 2002

- Simple algorithms

- Simple bump sensor

- No mapping of the room

- Random walk angle changing after bumping into an object or wall



Long exposure photo showing path taken by a Roomba as it cleans



机器人简史

• 2000 年代

○ 2004 - 2007 — DARPA Grand Challenge

- 首次自动驾驶竞赛
- CMU 于 2007 年获胜, 96 公里城市赛道用了 4 小时以上 (均速 22.53 km/h)



History of Robotics

• 2000s

○ 2004 - 2007 - DARPA Grand Challenge

- First autonomous driving competition
- CMU won in 2007, finishing a **96 km** urban course in over **4 hours** (avg. **22.53 km/h**)
 - So slow it was like a bicycle ride
- Already uses modern senses like cameras, lidars and radars*



* <http://www.tartanracing.org/tech.html>

* 论文: https://www.ri.cmu.edu/pub_files/2007/4/Tartan_Racing.pdf

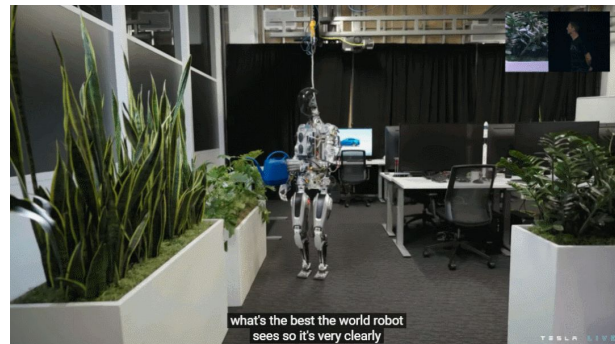
* <http://www.tartanracing.org/tech.html>

* paper: https://www.ri.cmu.edu/pub_files/2007/4/Tartan_Racing.pdf

今天的机器人

- 2010s - 2025

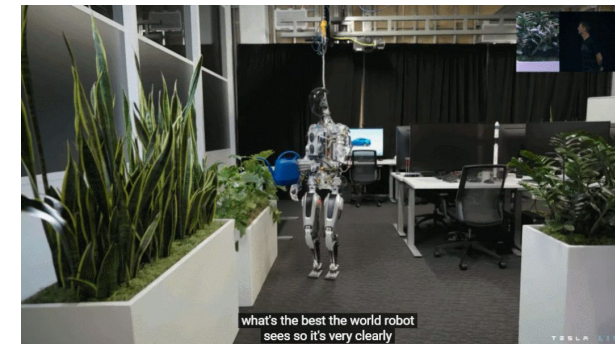
- 消费级无人机 — 大疆
- 自动驾驶汽车 — Waymo 等
- 高敏捷机器人 — Boston Dynamics
- 人形机器人 — Tesla 2022



Today's Robotics

- 2010s - 2025

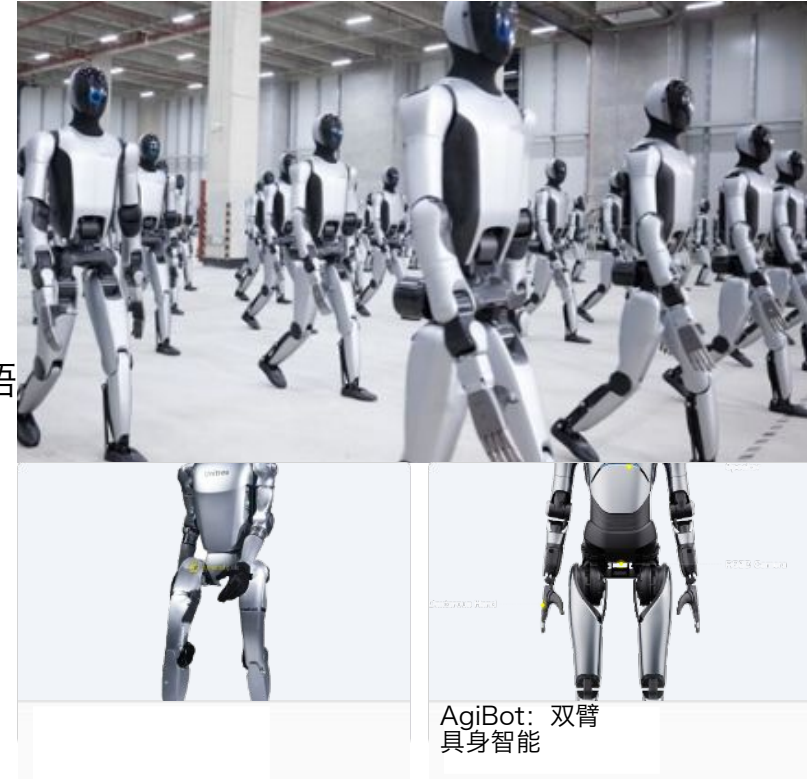
- Consumer-grade drones - DJI
- Self-driving cars - Waymo, etc.
- Highly-agile robots - Boston Dynamics
- Humanoid robot - Tesla 2022



今天的机器人

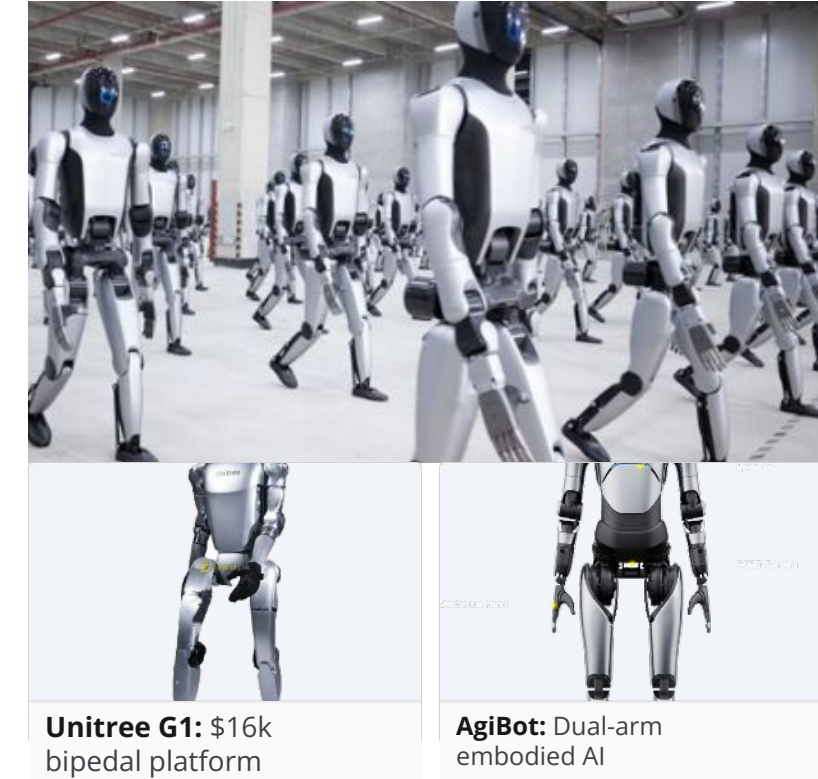
• 2025 - 2026: 中国人形机器人热潮

- 商业化与量产: Unitree (G1/R1) 把价格打到 16,000 美元以下, AgiBot 把硬件推向全球。
- 进入新能源工厂: 优必选 Walker S 在蔚来、比亚迪、富士康产线做真实工业试点。
- 由具身智能驱动: 从经典刚性控制转向多模态视觉-语言-动作 (VLA) 模型。
- 战略规模: 中国占全球人形专利 >50%, 2025/2026 估计产量超过 10,000 台。



Today's Robotics

- 2025 - 2026: Humanoid robot boom in China
 - Commercial & Mass Produced: Sub-\$16,000 price disruption by Unitree (G1/R1) and AgiBot scales hardware globally.
 - Deployed in EV Factories: Real-world industrial pilots by UBTECH (Walker S) in NIO, BYD, and Foxconn assembly plants.
 - Embodied AI Driven: Transitioning from classical rigid control to multimodal Vision-Language-Action (VLA) models.
 - Strategic Scale: **China accounts for >50% of global humanoid patents and an estimated 10,000+ units manufactured by 2025/2026.**



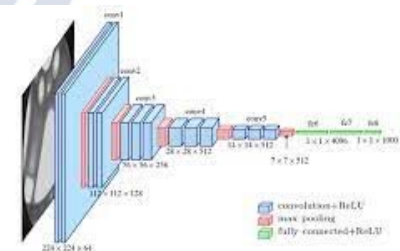
提纲

- 先介绍一下我
- 什么是具身智能?
- 机器人简史
- 为什么现在学具身智能?
- 课程内容概览
- 课程安排

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- **Why study Embodied AI now?**
- Course content overview
- Logistics

现在正是学习与从事具身智能的好时机



机器学习、语言与计算机视觉的近期突破: CLIP、ChatGPT。未来几年可期待更好的大规模视觉-语言模型

你的手机比 NASA 1969 年全部算力加起来还强数百万倍。

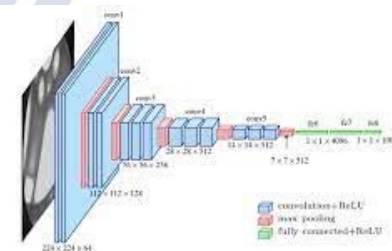
机器人硬件

更可靠、更便宜的机器人硬件。Boston Dynamics 的 Spot 约 75,000 美元, Unitree 有约 10,000 美元的产品

三维仿真器

更多开源、易用的三维仿真器。
*CARLA
*NVIDIA Isaac Sim

Now is a good time to study and work on Embodied AI



Artificial Intelligence

Recent breakthroughs in machine learning, language and computer vision: CLIP, ChatGPT. Expect a good Large Vision-Language Model in recent years

Computing Power

Your smartphone is millions of times more powerful than all of NASA's combined computing in 1969.

Robot Hardware

More reliable and affordable robot hardware. Boston Dynamics' spot costs \$75,000
Unitree has \$10,000 ones

3D Simulators

More open-source, easy-to-use 3D simulators.
*CARLA
*NVIDIA Isaac Sim

现在正是学习与从事具身智能的好时机

- 新冠让我们意识到：能减少人与人接触的具身智能很重要。



By Brianna Wessling | February 6, 2023

ABB uses robots to automate COVID antibody testing

The system, created by ABB and UTMB, can increase the number of neutralizing antibody tests performed from 15 a day to over 1,000 daily.

By Steve Crowe | January 8, 2021

Ava Robotics launches UV disinfection robot

Ava Robotics, the iRobot spinout known for its telepresence robots, is expanding its product portfolio with the introduction of an ultraviolet (UV) disinfection robot for corporate offices, warehouses, and other work spaces. The new robot is an evolution of a research project Ava Robotics conducted with the Massachusetts Institute of Technology and the Greater Boston...



9 robotics projects focusing on COVID-19

By Steve Crowe | September 24, 2021
Listen to this article

0:00 / 0:00

Voced by Amazon Polly



Now is a good time to study and work on Embodied AI

- Covid-19 makes us realize the importance of Embodied AI that limits human interactions.



By Brianna Wessling | February 6, 2023

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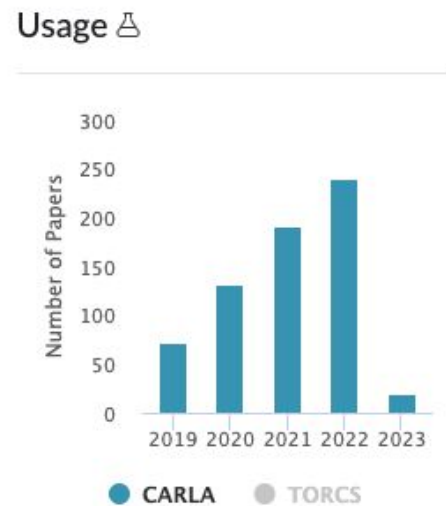
Voced by Amazon Polly



<https://www.therobotreport.com/tag/coronavirus/>

现在正是学习与从事具身智能的好时机

- 研究社区在快速增长
 - 顶会论文数量在上升!
 - ICRA、CoRL、RSS、IROS 等
 - 使用三维仿真器的论文迅速增加

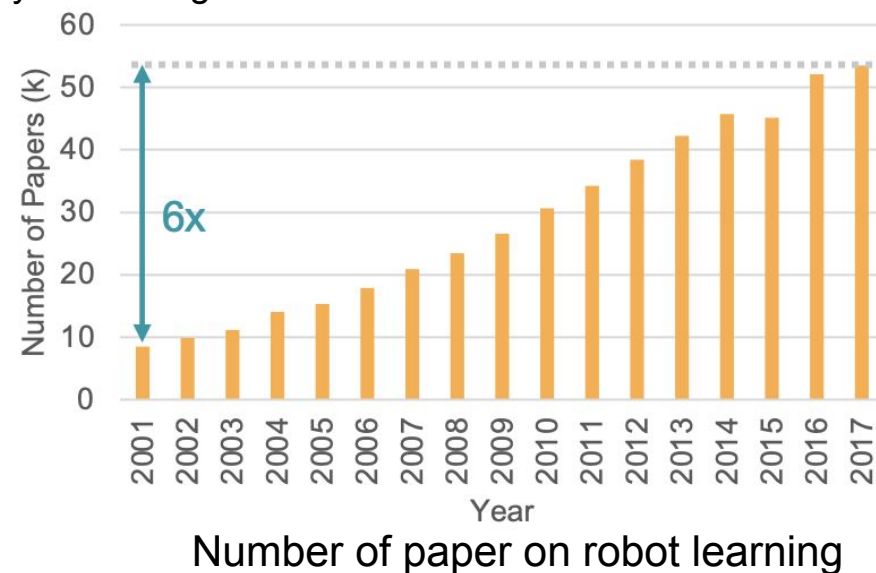
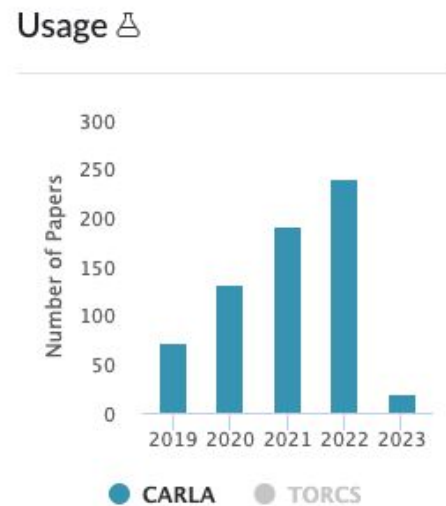


为什么多数时候会议比期刊更好?
因为研究进展很快……

Now is a good time to study and work on Embodied AI

- The research community is rapidly growing
 - Publications at top conferences are on the rise!
 - ICRA, CoRL, RSS, IROS, etc.
 - Papers using 3D simulator is rapidly increasing

Why conferences are better than journals (most of the time)?
Because the research happens fast...



具身智能很难！

来自 2015 年 DARPA Robotics Challenge

分三个阶段，开发机器人助手。任务包括：

1. 机器人开车通过障碍赛道到达目的地并下车
2. 从门进入建筑物
3. 转动阀门
4. 用工具在墙上开洞
5. 爬楼梯完成



只有 3 支队伍成功完成全部任务

Embodied AI is challenging!

From DARPA Robotics Challenge 2015

A 3-stage competition to develop robot assistants. Tasks involved:

1. Robot drives through an obstacle course to destination and exit car
2. Enters a building through door
3. Turns a valve
4. Cuts a hole in a wall using a tool
5. Climb stairs to finish



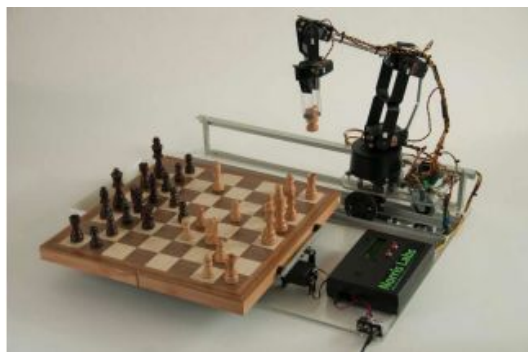
Only 3 teams successfully finished all the tasks

具身智能很难!

你可能在机器人文献里听过 “Moravec 悖论”

- Moravec 悖论 (1988) :

- “让计算机在智力测验或下棋上达到成人水平相对容易；但要让它们具备一岁儿童的感知与运动能力，却很难甚至不可能”



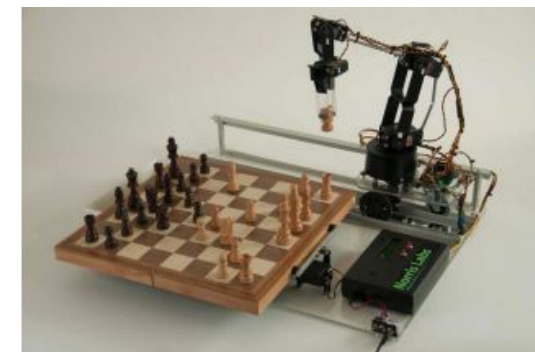
今天某种意义上仍成立。
你可以让 ChatGPT
写出动人故事，但它没法
帮你洗衣服

Embodied AI is challenging!

You may hear of “The Moravec's paradox” in the robotics literature

- The Moravec's paradox (1988):

- “it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”



Still true in some sense
nowadays. You can ask
ChatGPT to write a
convincing story but it
could not do your
laundry

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Boston Dynamics
拥有最先进的高敏捷机器人。
但短期内不会（以可负担价格）
出售。

Embodied AI is challenging!

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Boston Dynamics has the
most advanced highly-agile
robots. But they are not
selling it (and at an
affordable price) at any time
soon.

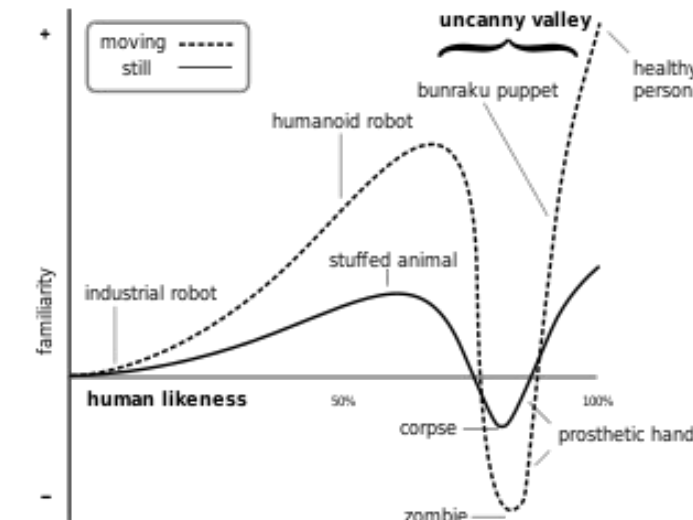
具身智能很难!

你可能也听过恐怖谷理论

恐怖谷理论:

- “……关于物体与人类相似程度，以及人们情绪反应之间的一种假说关系”
- 意思是：若要做人形机器人，必须做得非常好，否则人们会反感

到某个阶段，我们需要考虑人机交互的影响



当机器人越来越像人，熟悉感可能突然掉到负值

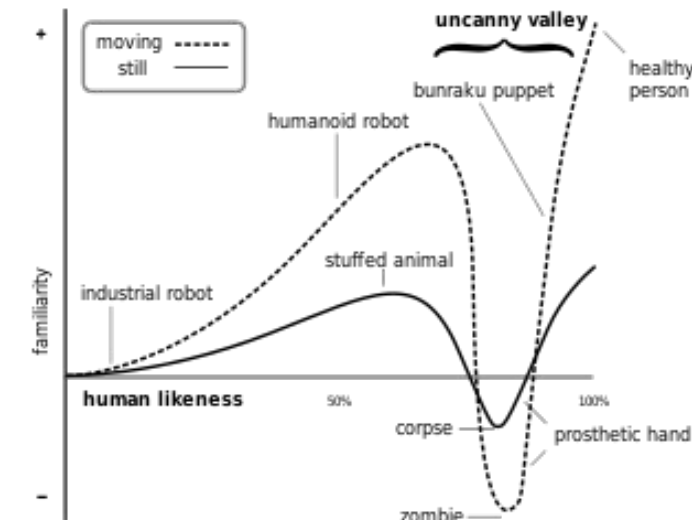


Embodied AI is challenging!

You may also hear of The uncanny valley (恐怖谷理论)

The uncanny valley (恐怖谷理论):

- “...a hypothesized relation between an object's degree of resemblance to a human being and the emotional response to the object”
- It means that if you want to build humanoid robot that looks like a human, it needs to be very good otherwise people will hate it
 - At some point we will need to consider the impact of human-computer interaction



As the robots become more like human, our familiarity with it may suddenly drops to a negative level



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具身智能的关键组成部分



观察并理解周围环境



机器人行为的
运动规划与控制



机器人系统

机器人本体。
三维仿真。
Sim2real

Key Components in Embodied AI



Perception

Seeing and understanding of
surroundings



Decision Making

Motion planning and
control of robot
behaviors



Robot Systems

Robot embodiment.
3D simulations.
Sim2real

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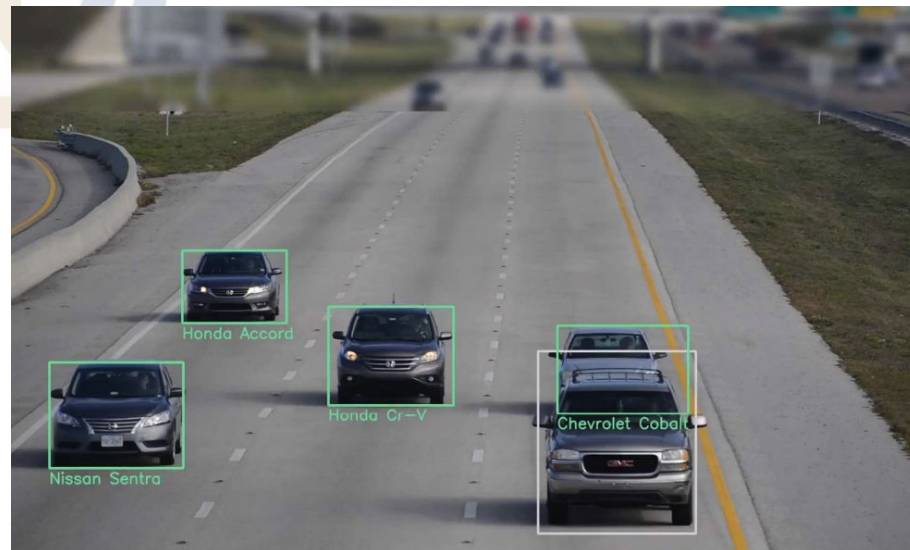


Robot Systems

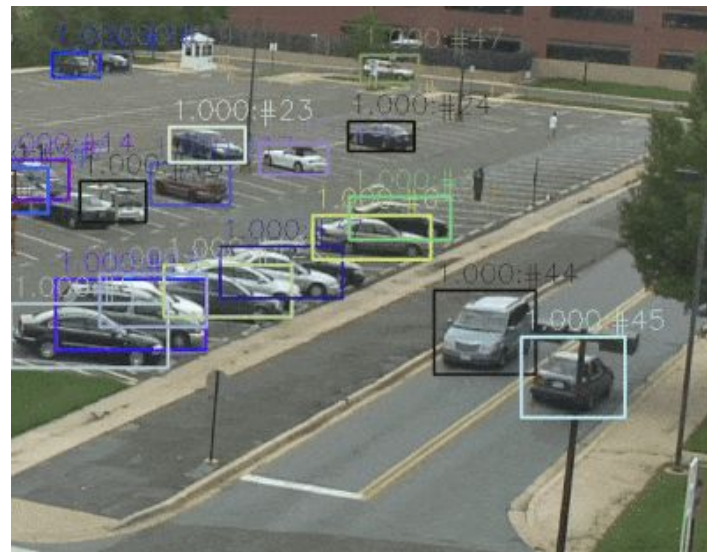
Robot embodiment.
3D simulations.
Sim2real

具身智能 — 感知

- 典型任务包括目标检测与跟踪



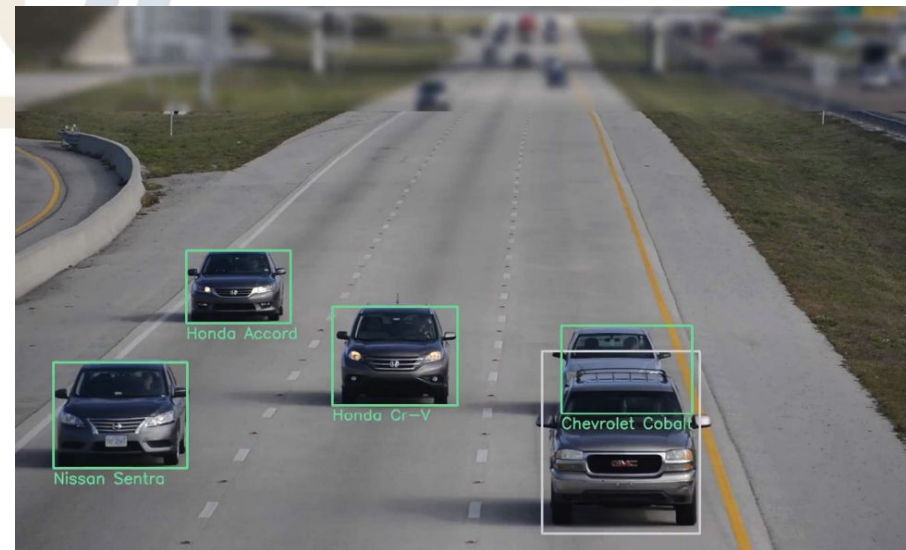
给不同类别物体画边界框



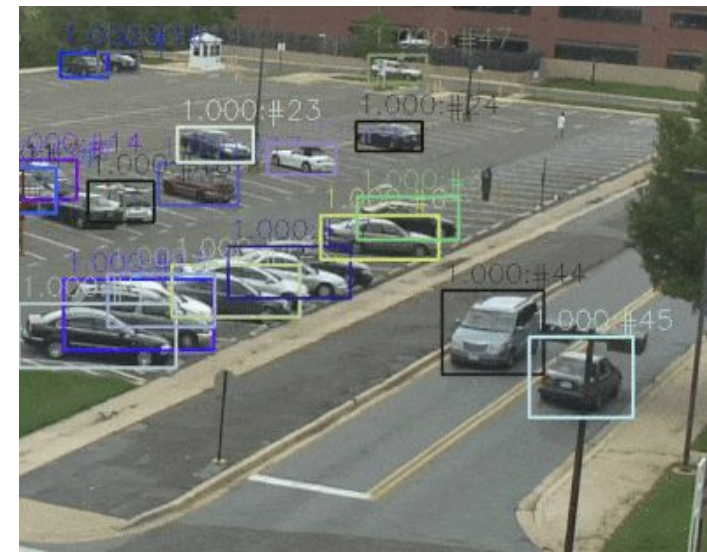
跟踪：为每个物体关联 ID

Embodied AI - Perception

- Typical tasks include Object detection & tracking



Drawing bounding boxes of different classes of objects

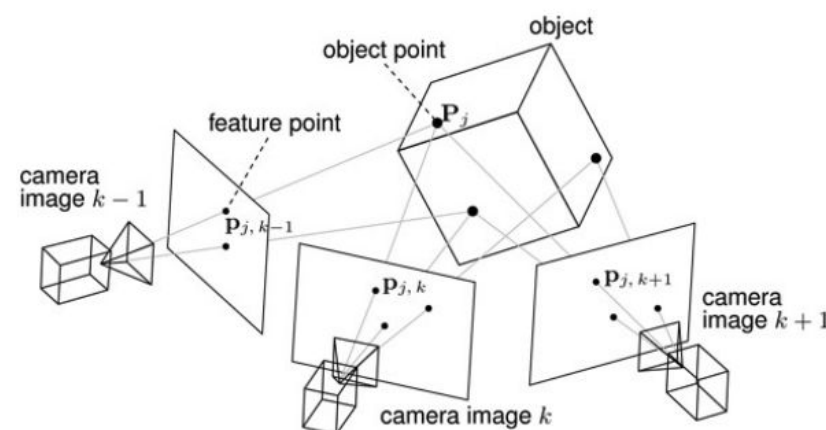
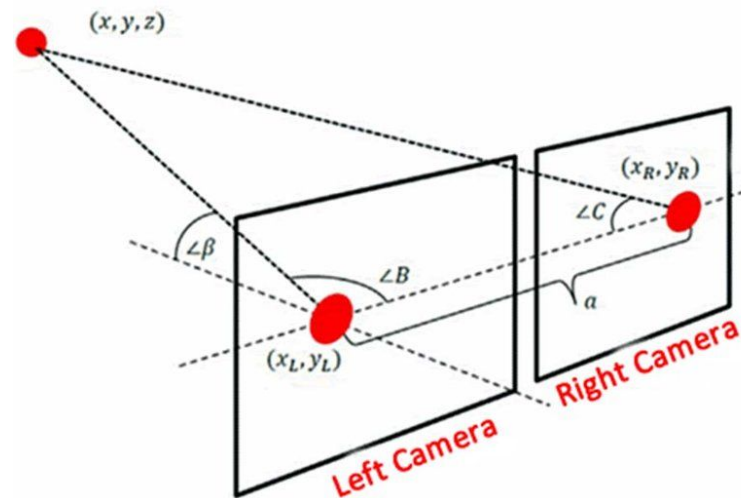


Tracking: Associate an ID for each object

具身智能 — 感知

• 立体视觉

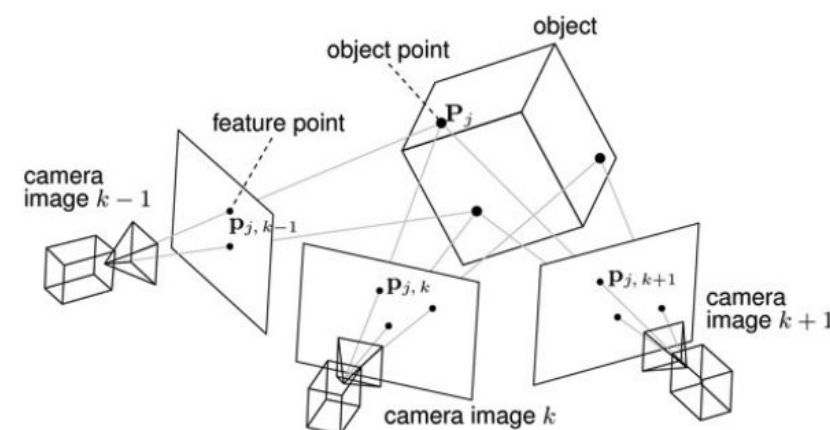
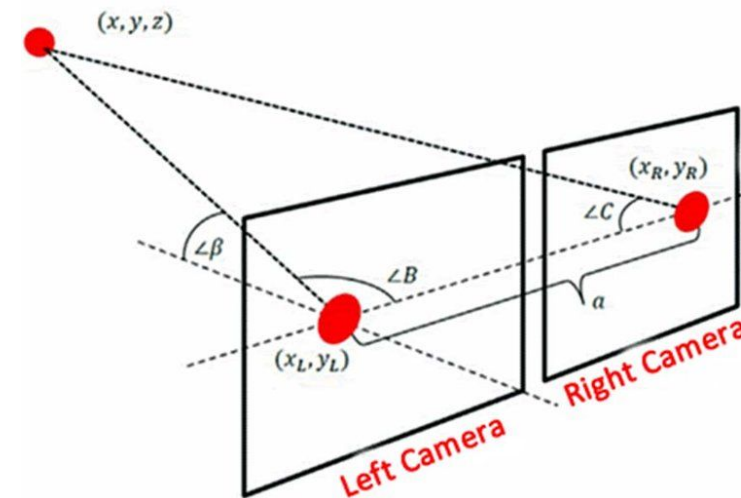
- 可用多相机几何估计深度（本课会讲）
- 也可用于三维重建



Embodied AI - Perception

• Stereo Vision

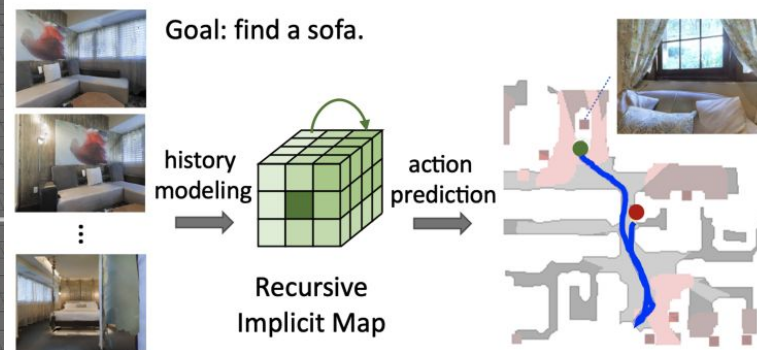
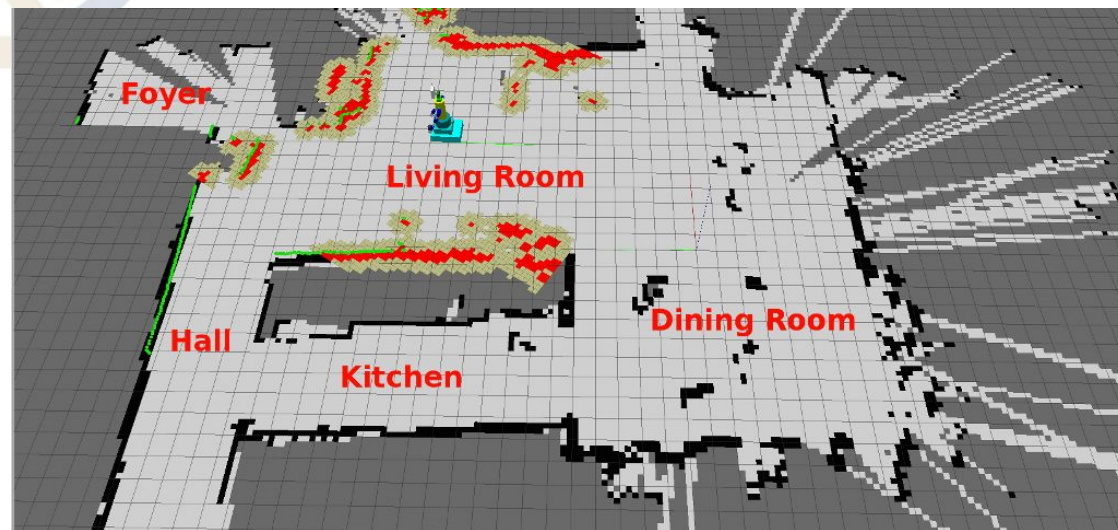
- Can be used to estimate depth using geometrics of multiple cameras (will cover in this course)
- And used for 3D reconstruction



具身智能 — 感知

• 定位与建图

- 对室内机器人很重要
- 可以像 SLAM 那样显式完成，也可以隐式完成

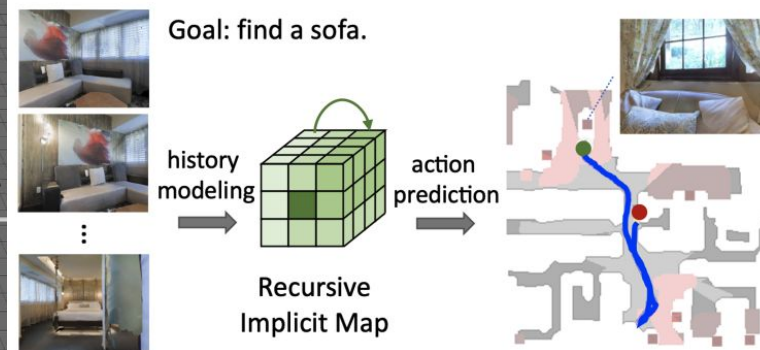
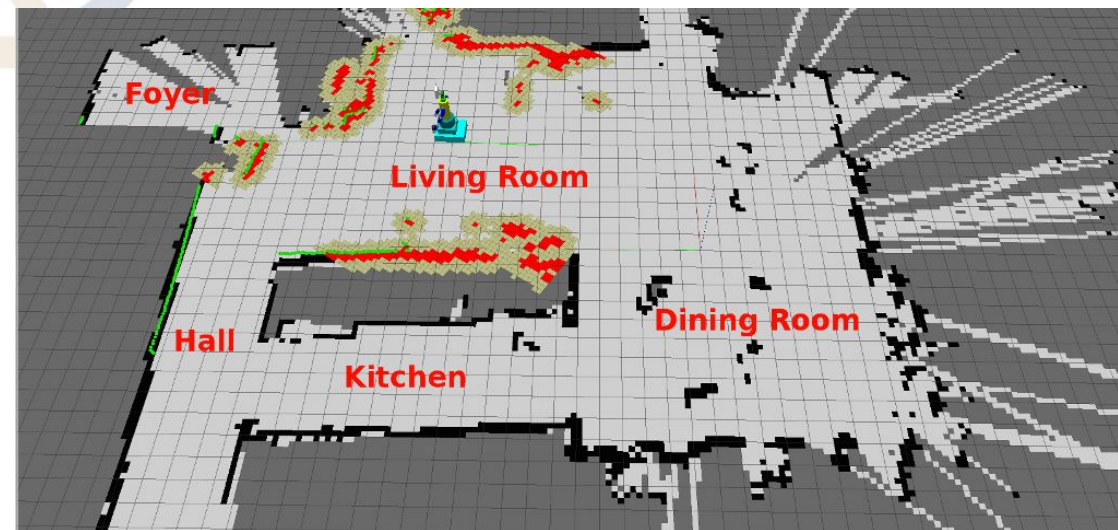


*Object Goal Navigation with Recursive Implicit Maps. IROS 2023

Embodied AI - Perception

• Localization and mapping

- Important for indoor robots
- Could be done explicitly like in SLAM or implicitly

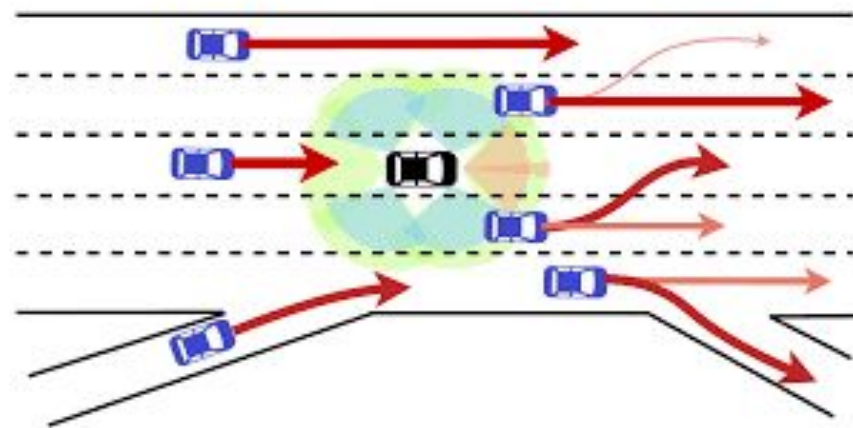
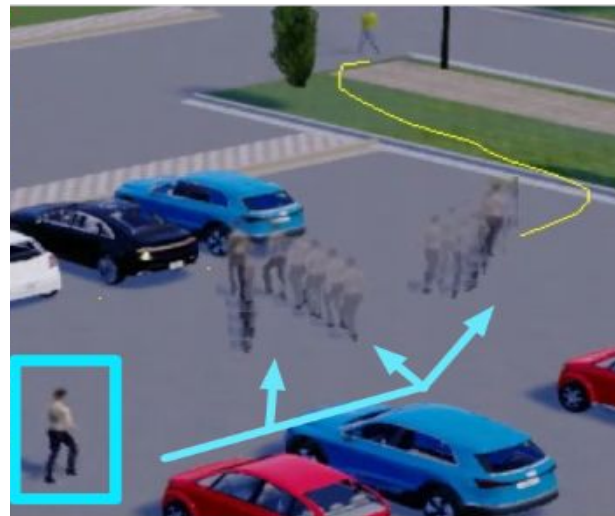


*Object Goal Navigation with Recursive Implicit Maps. IROS 2023

具身智能 — 感知

• 轨迹预测

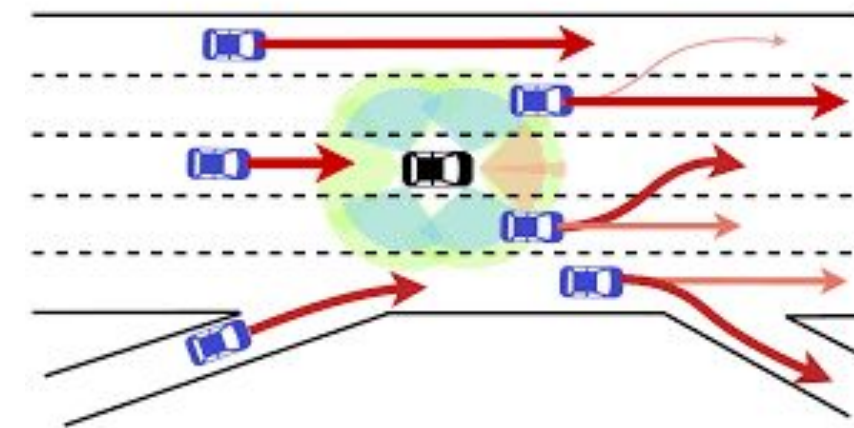
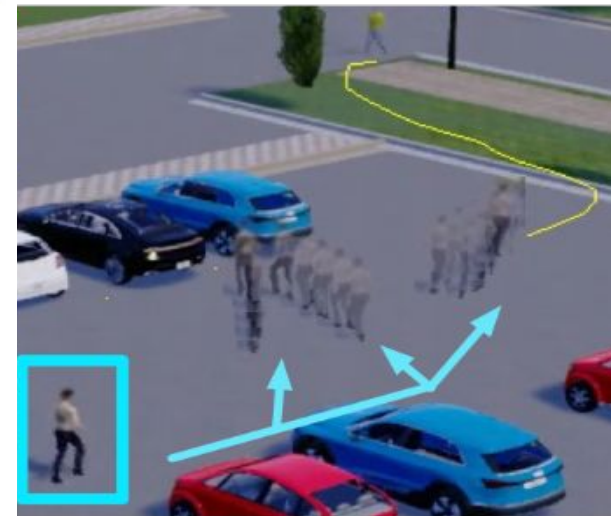
- 预测车辆与行人的未来轨迹/位置
- 通常预测未来 5 秒内



Embodied AI - Perception

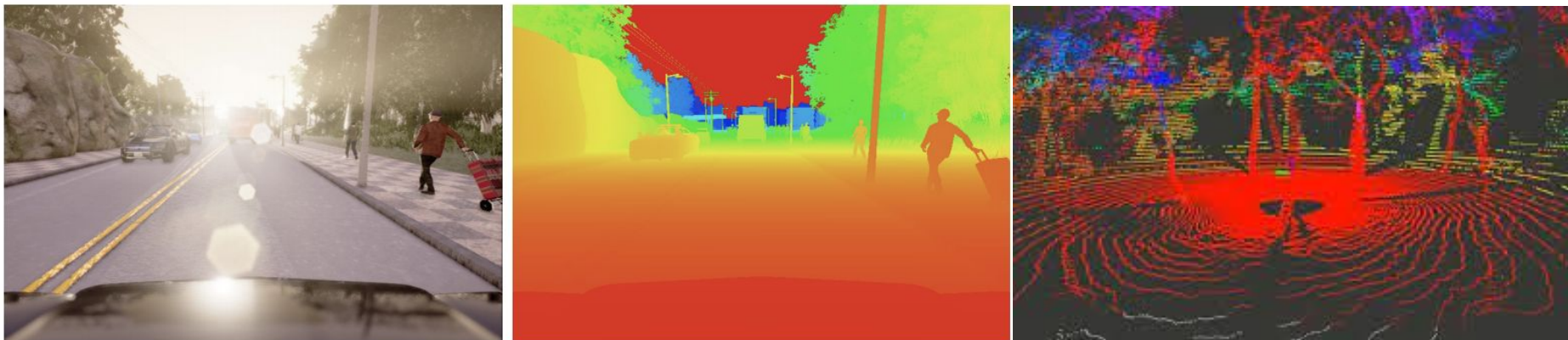
• Trajectory Prediction

- Predict the future trajectory/locations of vehicles and pedestrians
- Usually within 5 seconds in the future



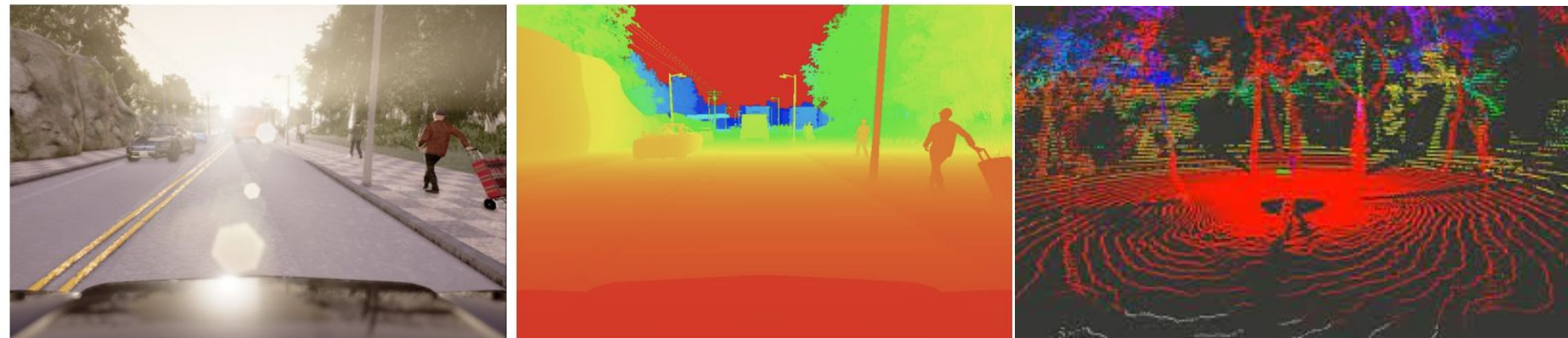
具身智能 — 感知

- 多模态传感器融合
 - 融合 RGB、深度与点云 (LiDAR) 信息



Embodied AI - Perception

- Multimodal Sensor Fusion
 - Combine information from RGB, Depth, and point clouds (LIDAR)



具身智能 — 感知

- 三维视觉定位 — SeeGround (CVPR 2025)
 - Grounding: 给定语言查询, 在三维场景中定位物体



Embodied AI - Perception

- 3D Visual Grounding - SeeGround (CVPR 2025)
 - Grounding: given language query, locate the object in the 3D scene



具身智能的关键组成部分



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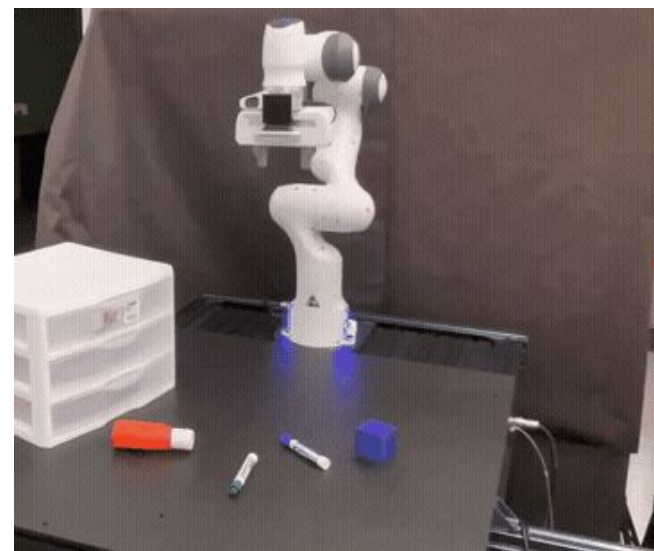


Robot Systems

Robot embodiment.
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具身智能 — 决策

- 运动规划与控制
 - 模型输出机器人要执行的未来运动计划



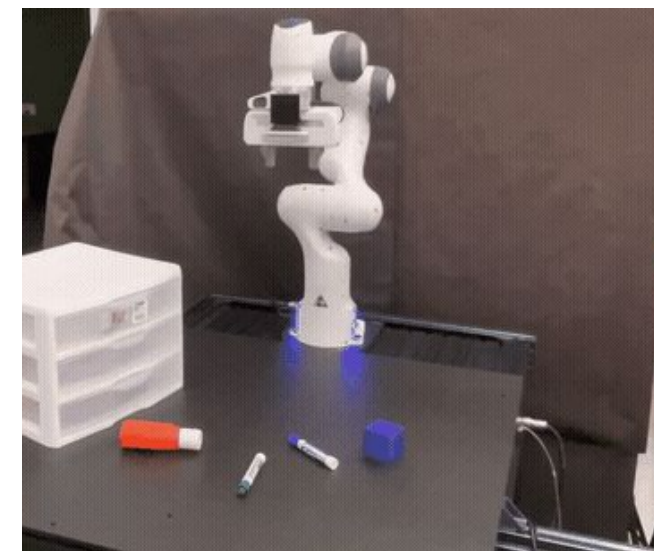
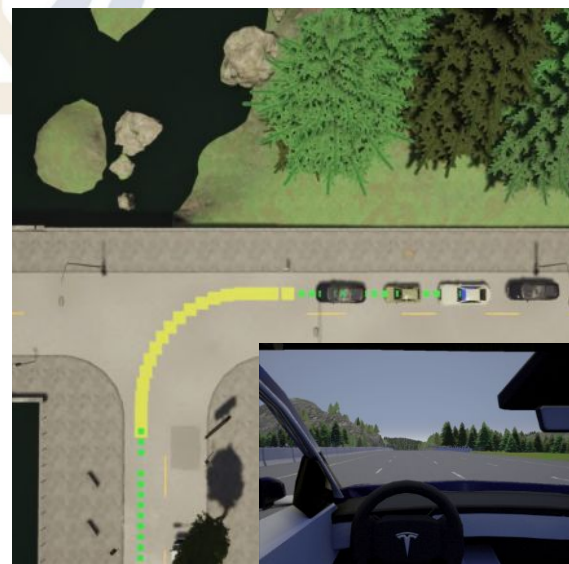
*RVT: Robotic View Transformer for 3D Object Manipulation. CoRL 2023



*Dit4Dit, 2026

Embodied AI - Decision Making

- Motion Planning and Control
 - Model outputs future motion plans for the robot to execute



*RVT: Robotic View Transformer for 3D Object Manipulation. CoRL 2023



*Dit4Dit, 2026

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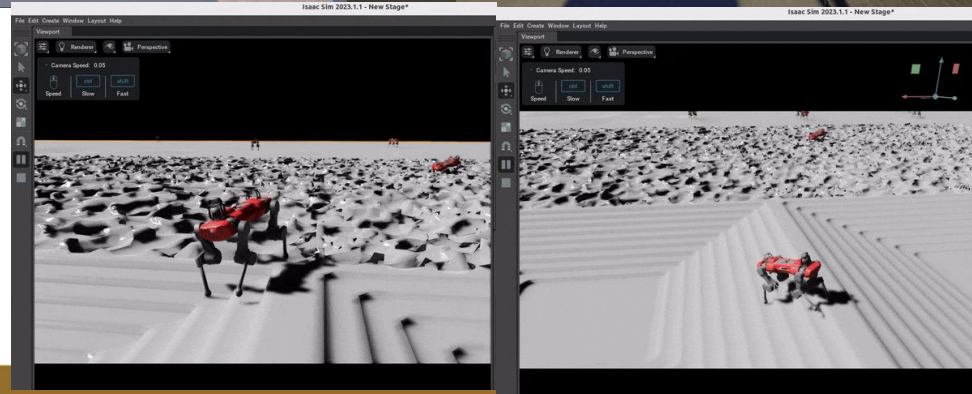


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具身智能 — 机器人系统

- 三维仿真是机器人研究的重要部分

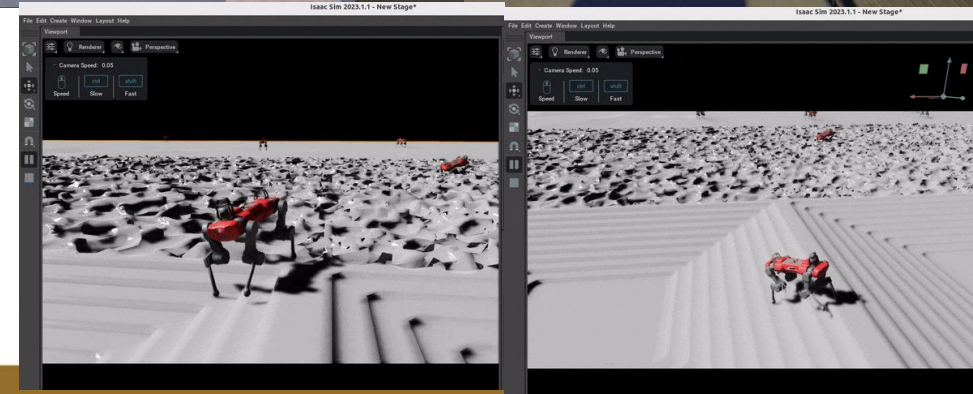


* CARLA 仿真器
(基于 UE4):
<https://carla.org/>
<https://leaderboard.carla.org/>

* Nvidia Isaac Sim:
<https://developer.nvidia.com/isaac-sim>

Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research



* CARLA simulator
(based on UE4):
<https://carla.org/>
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具身智能 — 机器人系统

三维仿真器
可以非常真实

这来自
CARLA —
专为自动驾
驶任务设计



Embodied AI - Robot Systems

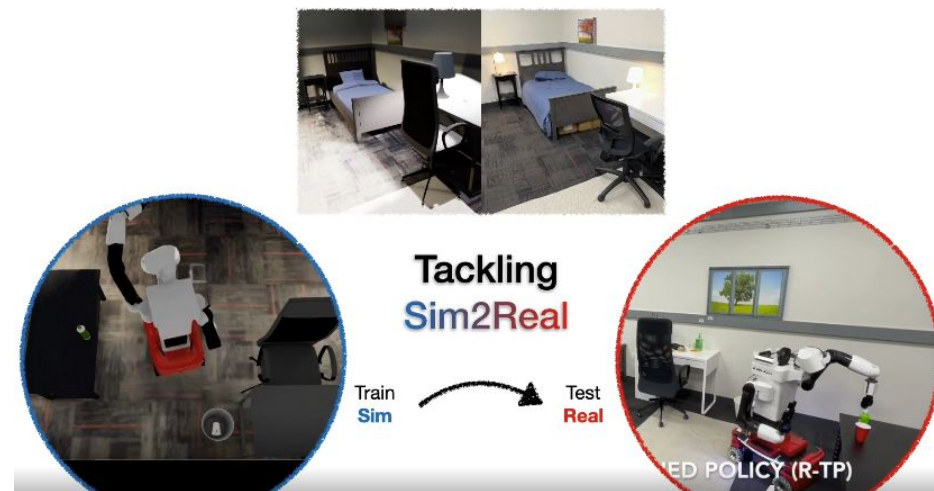
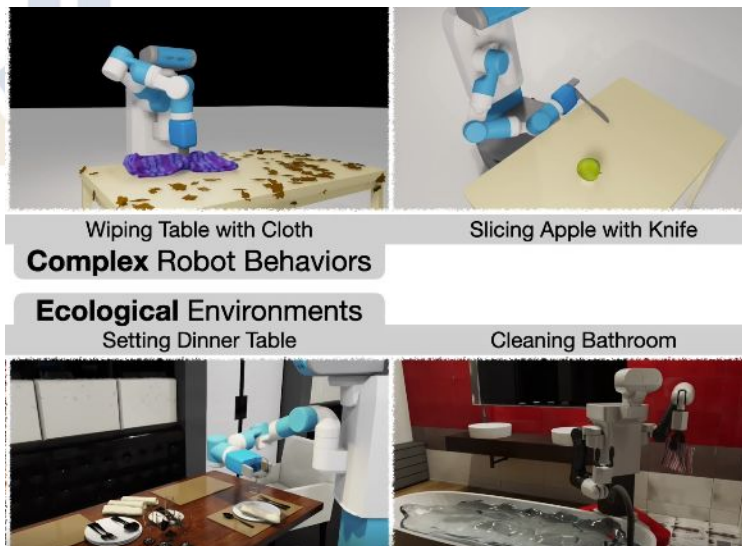
3D simulator
can be very
realistic

This is from
CARLA -
Specialized
for
self-driving
task



具身智能 — 机器人系统

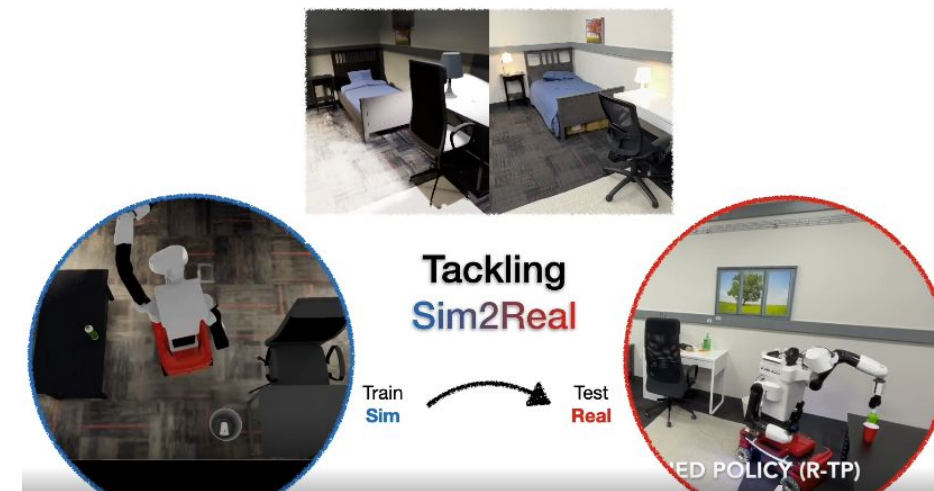
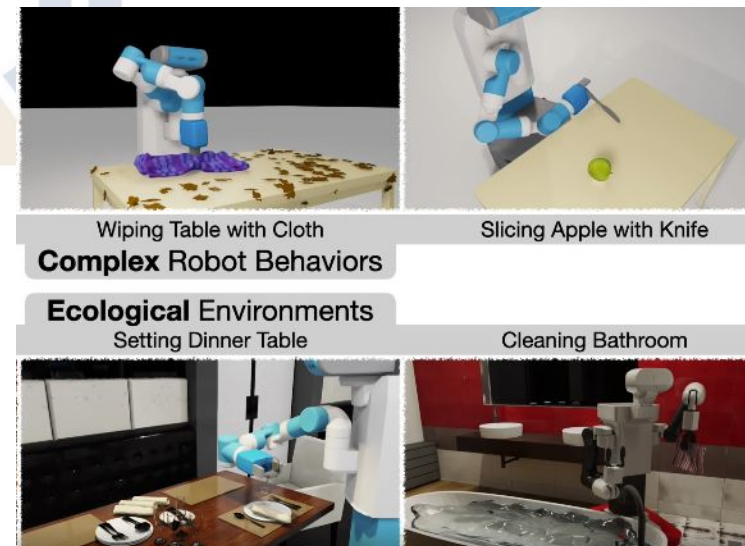
- 三维仿真是机器人研究的重要部分



* BEHAVIOR-1K (基于 Isaac Sim) : <https://behavior.stanford.edu/behavior-1k>

Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research

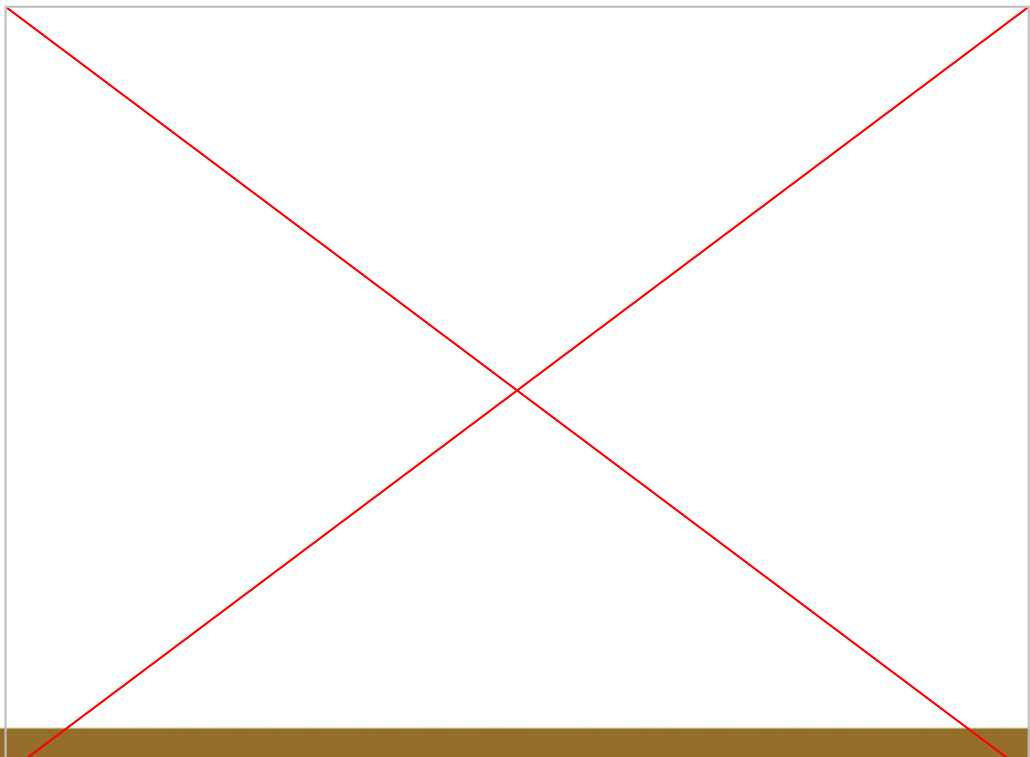


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具身智能 — 机器人系统

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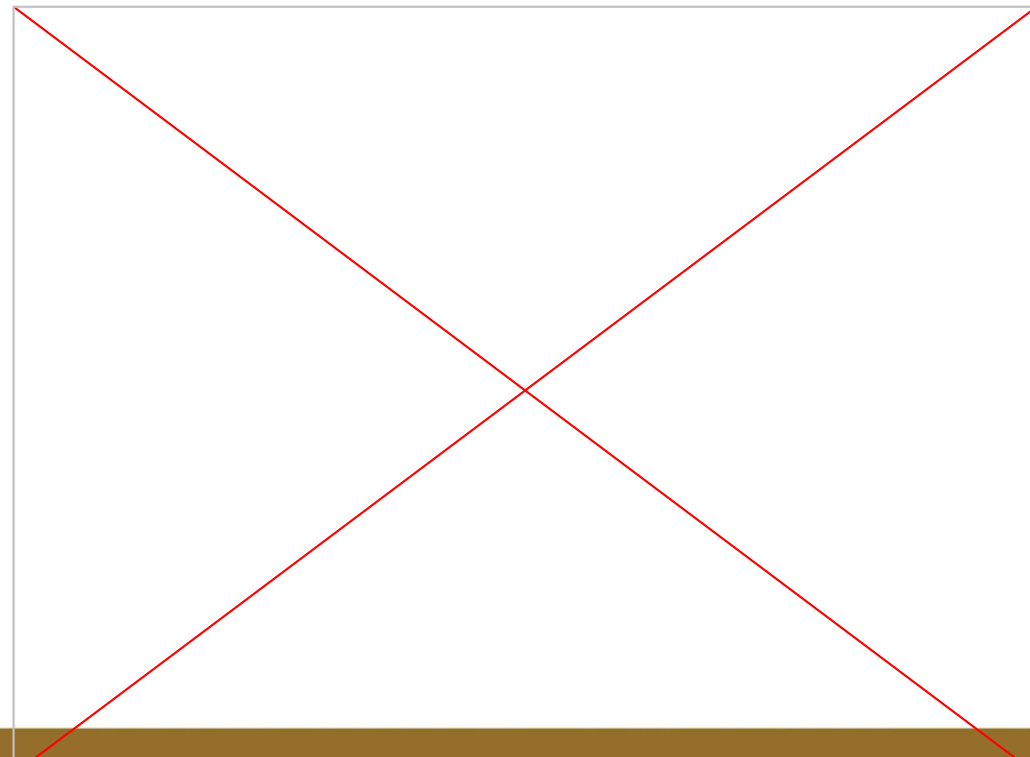
这来自 BEHAVIOR-1k —
专为日常任务机器人设计



Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research

This is from BEHAVIOR-1k -
Specialized for everyday task
robots



具身智能的关键组成部分



观察并理解周围环境



机器人行为的
运动规划与控制



机器人系统

机器人本体。
三维仿真。
Sim2real

Key Components in Embodied AI



Perception

Seeing and understanding of
surroundings



Decision Making

Motion planning and
control of robot
behaviors



Robot Systems

Robot embodiment.
3D simulations.
Sim2real

课程计划

- 机器感知
 - 几何计算机视觉
 - 视觉识别
 - 定位与建图
- 机器人系统
 - ROS
 - 仿真
- 决策
 - 规划与控制
 - 状态估计、运动学与动力学
 - 具身导航
 - 马尔可夫决策过程、强化学习等
 - 具身交互与操作
 - VLA 等

Course Plan

- Machine Perception
 - Geometric computer vision
 - Visual Recognition
 - Localization and Mapping
- Robot Systems
 - ROS
 - Simulation
- Decision Making
 - Planning and Control
 - State estimation, Kinematics and dynamics
 - **Embodied Navigation**
 - **Markov Decision Process, Reinforcement Learning, etc.**
 - **Embodied Interaction and Manipulation**
 - **VLA, etc.**

课程目标

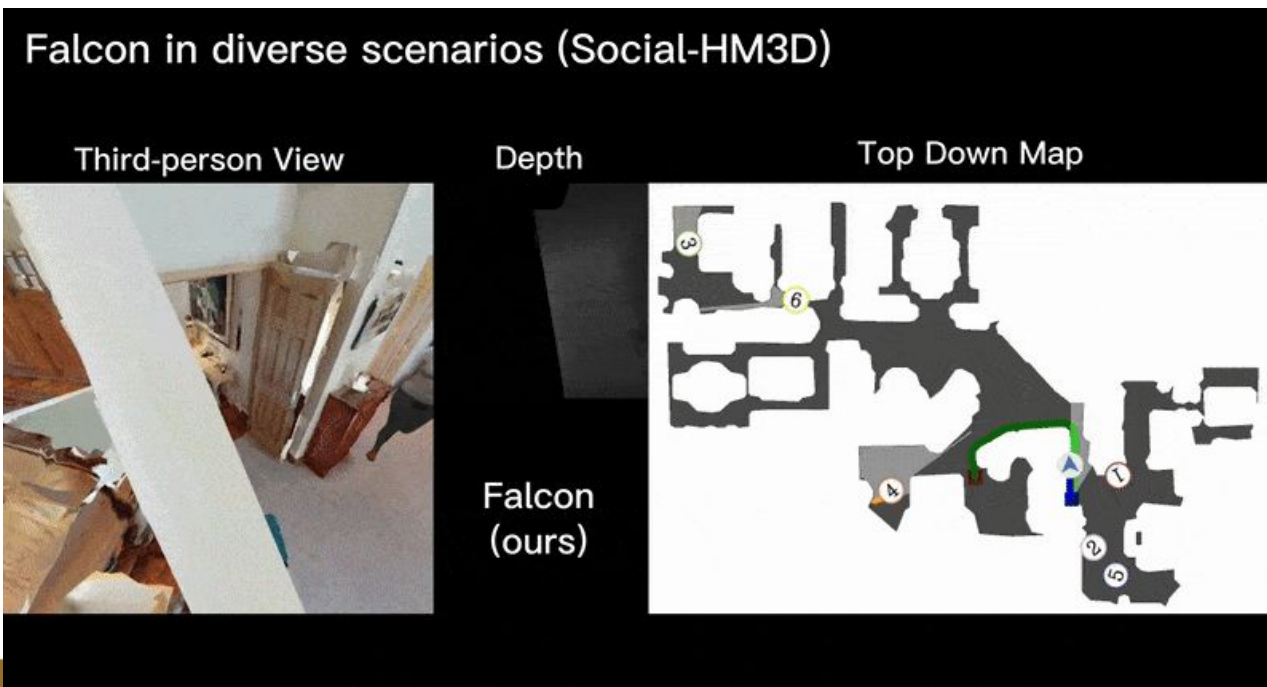
- 描述具身智能中的关键原理、概念、模型与技术
 - 理解技术挑战与近期研究进展
- 实现课上介绍的算法，解决小型真实问题
 - 熟悉 ROS、三维仿真器等具身智能系统
- 为具身智能应用发展新方法
 - 导航
 - 操作

Course Objectives

- Describe key principles, concepts, models and techniques in Embodied AI
 - Understand technical challenges and recent research progress
- Implement algorithms introduced in class to solve small real-world problems
 - Familiarity with Embodied AI systems like ROS and 3D simulators
- Develop novel methods for Embodied AI applications
 - Navigation
 - Manipulation

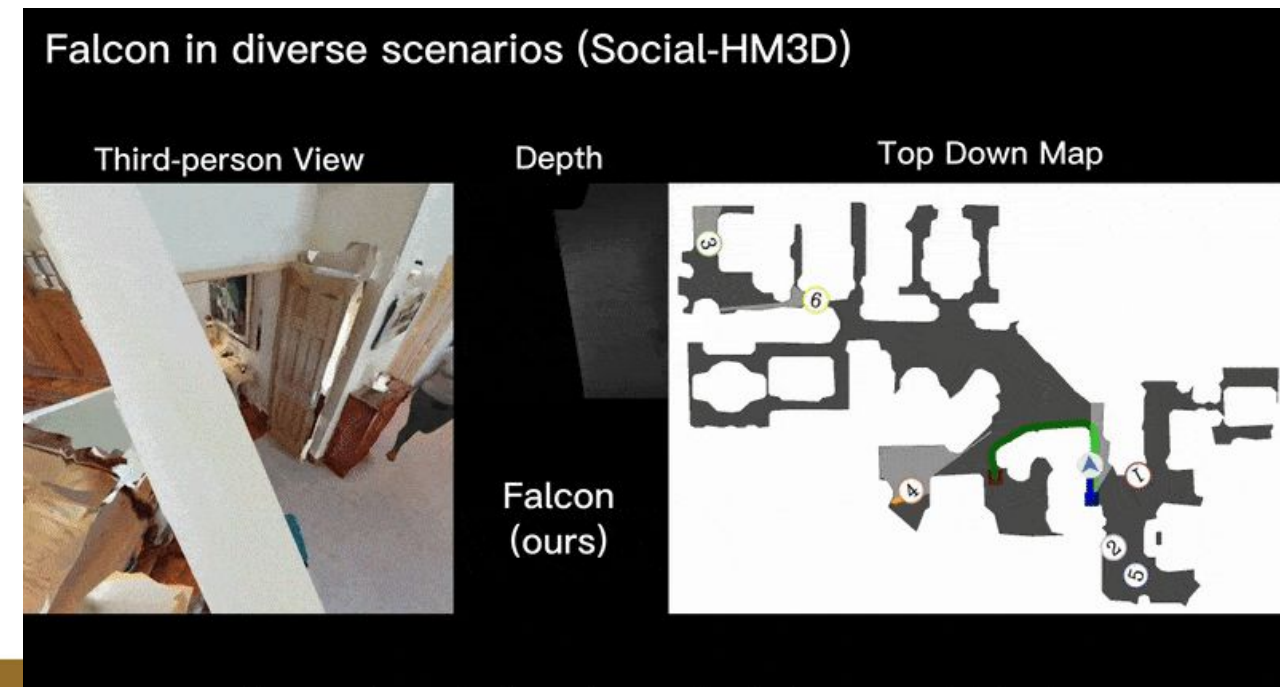
课程目标

- 为具身智能应用发展新方法
 - 课程结束时，你能做出能避开人的导航机器狗！



Course Objectives

- Develop novel methods for Embodied AI applications
 - By the end of the course you can build your navigation robot dog that can avoid humans!



提纲

- 先介绍一下我
- 什么是具身智能？
- 机器人简史
- 为什么现在学具身智能？
- 课程内容概览
- 课程安排

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- **Logistics**

课程安排：第1部分

- 课程网站
 - <https://hkust-aiaa4220.github.io/fall2026/lecs.html>
 - 可查看课程日程、作业链接、课件下载、课程政策等
 - 课件会在每次上课前发布

Embodied AI

AIAA 4220, Fall 2026

Instructors: [Junwei Liang](#)



Home		Schedule
Lecture:	Date and Time: 09:00AM - 11:50AM, Monday	
	Location: Rm 233, W1	
	Websites: Instructure	
	Zoom: 628-334-1826 PW: 12345 (In-person expected) (Recordings will be available after each class)	
Course Description:	This course aims to provide students with key principles and algorithms to build modern Embodied AI systems. Key topics include machine perception, embodied navigation and manipulation algorithms. Through this course, students will learn and practice the foundational principles, techniques, and tools to build new Embodied AI systems.	
Target Audience/ Prerequisites:	This is a course primarily for junior and senior students.	
Course Work:	Grading	Grading will be based on best 5 of 6 in-class quizzes (25%), one paper presentation (25%), one machine perception homework (25%) and one simulation project (25%). • The in-class quizzes are designed to complement the lectures and making sure you understand the key knowledge about the course. • There are 6 in-class quizzes over the semester. We will take your best 5 out of the 6 quizzes as your grade.
	In-Class Quizzes	

Logistics: Part 1

- Course website
 - <https://hkust-aiaa4220.github.io/fall2026/lecs.html>
 - You can see the **course schedule**, **homework links**, **download slides**, **policy**, etc.
 - The slides will be available before each class

Embodied AI

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	In-Class Quizzes	

课程安排：第1部分

- Canvas
 - <https://hkust-gz.instructure.com/courses/3918>
 - 课程通知会发在这里
 - 问题与讨论请用讨论区
 - 随堂测验也在 Canvas 上进行
- 计算资源
 - 项目 2 会给每人 200 元大学 HPC 额度
 - 由 AI Thrust 提供
 - 9/14 之后发放账号密码
 - 阿里云: <https://university.aliyun.com/> 免费 300 元额度
 - * 项目 3 有 5 个仿真题目可选
 - 大多数不需要 GPU

Logistics: Part 1

- Canvas
 - <https://hkust-gz.instructure.com/courses/3918>
 - Course announcement will be posted here
 - Use the discussion section for questions and discussions
 - **We will use Canvas for in-class quizzes**
- Compute resources
 - We will provide you each with a ¥200 credit for university HPC for project 2
 - Courtesy of AI Thrust
 - We will distribute the account/password after 9/14
 - Aliyun 阿里云: <https://university.aliyun.com/> free ¥300 RMB credit
 - * There will be 5 simulation projects you can choose from for Project 3
 - Most of them does not require a GPU

课程安排 — 阿里云计算资源

- 阿里云为我校提供了免费或优惠计算资源

- <https://university.aliyun.com/>
- 300 元代金券
- 每年最高 5000 元订单 7 折
- 可用港科广邮箱直接申请



The screenshot shows the Aliyun University Cloud Resource page. At the top, there is a banner with the text "云工开物" and "阿里云高校计划, 助力高校科研与教育加速, 让每位中国在校大学生真实受益于普惠算力". Below the banner, there is a section titled "助力高校师生云上‘创世界’" with a quote from the book "天工开物". The main content area is divided into two tabs: "面向高校学生" (Selected) and "面向高校教师". Under the "面向高校学生" tab, there are three sections: "高校学生权益" (General), "高校学生通用权益" (General), and "合作高校学生专属权益" (Exclusive). Each section includes a description of the benefits, a "立即领取" (Get It Now) button, and a "适用范围" (Applicable Scope) section.

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助力高校师生云上“创世界”

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面向高校学生 | 面向高校教师

高校学生权益
面向中国高校学生, 立即领取权益开启云上实践 [活动规则](#)

高校学生通用权益
完成高校学生认证即可领取, 其中港澳台地区近期开启

300
无门槛优惠券

适用范围: 阿里云全量公共云产品 (特殊商品除外)
使用说明: 自领取之日起, 1年内有效, 如订单退款, 优惠券不

合作高校学生专属权益
[合作高校](#) 学生完成学生认证即可领取, 可与300元优惠券叠加使用

3折
折扣优惠

适用范围: 阿里云全量公共云产品 (特殊商品除外), 封顶5000元/年
使用说明: 自领取之日起, 1年内有效, 如订单退款, 仍占用优

Logistics - Computation Resources from Aliyun

- Aliyun has provided our university with free or discounted computation resources
 - <https://university.aliyun.com/>
 - 300 RMB voucher
 - 70% off for orders up to 5000 RMB/year
 - You can directly apply with your hkust-gz email



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助力高校师生云上“创世界”

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面向高校学生 | 面向高校教师

高校学生权益
面向中国高校学生, 立即领取权益开启云上实践 [活动规则](#)

高校学生通用权益
完成高校学生认证即可领取, 其中港澳台地区近期开启

300
无门槛优惠券

适用范围: 阿里云全量公共云产品 (特殊商品除外)
使用说明: 自领取之日起, 1年内有效, 如订单退款, 优惠券不

合作高校学生专属权益
[合作高校](#) 学生完成学生认证即可领取, 可与300元优惠券叠加使用

3折
折扣优惠

适用范围: 阿里云全量公共云产品 (特殊商品除外), 封顶5000元/年
使用说明: 自领取之日起, 1年内有效, 如订单退款, 仍占用优

课程安排：第2部分

- 课程强调动手实践
 - 意味着编程与实验
- 先修要求
 - Python、基础微积分、线性代数、概率论
- 强烈建议用 Ubuntu（不要用 Windows）— 多数代码/工具基于它
 - 语言：Python
 - 工具：PyTorch / Habitat Sim / Mujoco / Isaac Sim
- 欢迎使用其他语言/工具，但助教可能无法提供帮助
 - 课堂参与
- 鼓励举手提问打断
- 之后在 Canvas（或课堂上）提问也很好
- 互动能提升学习（课堂与办公时间都是）
 - 办公时间
 - E4 304 室
 - 周三 10:00 - 11:30（请先发邮件预约）

Logistics: Part 2

- Course aims to give you more hands-on experience
 - Means coding and experimenting
- Prerequisites
 - Python programming, Basic calculus, linear algebra, probability theory
 - You should really be using Ubuntu (not Windows) - Most codes/tools are based on that
- Language of choice: Python
- Toolkit of choice: PyTorch / Habitat Sim / Mujoco / Isaac Sim
 - You are welcome to use other languages/toolkits, but the TAs will not be able to help
- Participation
 - Interrupting (by raising a hand) to ask your question is strongly encouraged
 - Asking questions later on Canvas (or in class) is also great
 - Interaction improves learning (both in-class and at my office hours)
- Office hours
 - Rm 304, E4
 - Wednesday 10:00AM - 11:30AM (Please email me first to make an appointment)

评分

- A 档 (A+、A、A-)：通常要求总分至少 80，且排名前 35%;
- A+：通常要求总分至少 95，全课程表现突出，且排名前 10%
- B 档 (B+、B、B-)：通常要求总分至少 70。
- C 档 (C+、C、C-)：通常要求总分至少 60。
- D：通常要求总分至少 50。
- F：总分低于 50。

以上评分规则可能随学校规定和成绩分布调整。

Grading

- A range (A+, A, A-): normally requires an overall course grade of at least 80 AND Ranked in the top 35%;
 - A+: normally requires an overall course grade of at least 95 and exceptional performance across the course AND Ranked in the top 10%
- B range (B+, B, B-): normally requires an overall course grade of at least 70.
- C range (C+, C, C-): normally requires an overall course grade of at least 60.
- D: normally requires an overall course grade of at least 50.
- F: normally applies to an overall course grade below 50.

The above grading rules may be subject to modifications depending on the university regulations and the actual distributions of the course grades.

评分 — 总览

- 课程评价构成

- (25%) 6 次随堂测验，选择题（取 6 次中最好的 5 次）
- (25%) 项目 1：课堂论文报告 — 个人作业
- (25%) 项目 2：第一人称目标检测 — 个人作业
- (25%) 项目 3：仿真项目 — 2 - 3 人小组
- 不记考勤！但随堂测验和论文报告你必须到场

Grading - Overview

- Course evaluation overview

- (25%) 6 in-class quizzes, multiple-choice questionnaires (**Grading: best 5 out of 6**)
- (25%) Project 1: In-class Paper Presentation - Single-person work
- (25%) Project 2: Egocentric Object Detection - Single-person work
- (25%) Project 3: Simulation Project - Group of 2-3 people
- We do not take attendance! But you should be here for the in-class quiz and your paper presentation

评分 — 细节

- 随堂测验 (25%)

- 配合讲座，确认你掌握课程关键知识
- 6 次个人测验（必须自己完成，不允许用 LLM）

- 按最好的 5/6 计分（所以可以缺一次仍拿满分）

- 发布在 Canvas Quizzes

- 时间安排

- 10:10 - 10:20

- 测验 1: 9 月 21 日

- 测验 2: 10 月 12 日

- 测验 3: 10 月 26 日

- 测验 4: 11 月 9 日

- 测验 5: 11 月 23 日

- 测验 6: 11 月 30 日

Grading - Details

- In-class Quizzes (25%)

- Designed to complement the lectures and making sure you understand the key knowledge about the course
- 6 individual assignment (so you need to complete this yourselves and no LLM allowed)
 - Grading based on your best 5 out of 6 (so you can miss one and still get full 100%)
- **Will be posted on Canvas Quizzes section**
- Schedule
 - 10:10 - 10:20 am.
 - Test 1: Sep 21
 - Test 2: Oct 12
 - Test 3: Oct 26
 - Test 4: Nov 9
 - Test 5: Nov 23
 - Test 6: Nov 30

评分 — 细节

- 项目 1：论文报告（25%）

- 从具身智能论文列表中选 1 篇报告。要证明你理解论文，并能讲给不熟悉该文的同学。10 分钟报告 + 5 分钟至少回答一个问题（来自助教/同学/老师）

- 论文列表: [link](#)

- 完整日程

- P1 报告顺序 9/21 公布
 - P1 报告 #1 - 8: 10/12 至 11/30

- （每次课 5 位同学）（从 10:10 开始）

- 这是个人作业

- 项目 1 占 25%

- 评分: 报告 50% + 幻灯片 50%，由老师和助教评定。请使用模板: [link](#)

- 报告后需提交幻灯片

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Grading - Details

- Project 1: Paper Presentation (25%)

- You will select 1 paper from our list of Embodied AI research to present. This should show that you understand the paper, and you could present it to students who do not know the paper well.

- You have 10 minutes to present and 5 minutes to handle at least one question (from TA/peers/teacher)**

- Paper list: [link](#)

- Full schedule

- P1 presentation order release on 9/21
 - P1 presentation #1-8 from 10/12 to 11/30
 - (5 students per lecture) (starting at 10:10 am.)

- This is an individual assignment

- Project 1 is worth 25% credit

- The grading is based on both the presentation (50%) and the slides (50%), by the teacher and TAs. Please follow the slide template here: [link](#)
 - You are required to submit the presentation slides after the presentation

评分 — 细节

- 项目 2: 机器感知作业 (25%)

- 第一人称视角目标检测作业

- 个人作业

- 提交到 Kaggle 排行榜。

- 按模型绝对分数和项目报告评分

- 详细评分标准: [link](#)

- 将公布三个绝对分数阈值

- S: 满分

- A: 80%

- B: 70%

- 暂定日程:

- P2 于 9/21 发布

- Kaggle 提交截止 10/26

- 报告截止 10/30

Grading - Details

- Project 2: Machine Perception Homework (25%)

- You will do an Ego-view object detection homework

- This is an individual assignment

- You will submit to a Kaggle leaderboard.

- **Grading based on model absolute score and project report**

- Detailed grading rubrics: [link](#)

- Three absolute score thresholds will be published

- S: full credit / A: 80% / B: 70%

- Tentative schedule:

- P2 release on 9/21

- Kaggle submission due on 10/26

- Report due on 10/30

评分 — 细节

- 项目 3: 仿真项目 (25%) — 2 - 3 人小组完成

- 可从 5 条赛道中选择:

- MiniGrid: 稀疏奖励导航
 - BabyAI: 语言引导模仿学习
 - MuJoCo: 鲁棒连续控制
 - ManiSkill: 桌面操作
 - Habitat: 避障导航

- 任务: 复现基线, 提出一个有动机的改动, 并通过公平对照实验评估。分析成功与失败。

- 交付: 15 分钟报告 (12+3 问答) 和报告

- 评分: 报告 50% + 文稿 50%。看重扎实实验与清晰分析; 不必超过基线。

- 暂定日程:

- P3 手册 10/19 发布
 - 11/2 前组队
 - 12/7 报告; 报告与幻灯片 12/20 截止

Grading - Details

- Project 3: Simulation Project (25%) - Done in groups of 2-3 people

- You can choose from 5 tracks:

- MiniGrid: Sparse-reward navigation
 - BabyAI: Language-guided imitation learning
 - MuJoCo: Robust continuous control
 - ManiSkill: Tabletop manipulation
 - Habitat: Collision-aware navigation

- Task: Reproduce a baseline, introduce one motivated modification, and evaluate it through a fair, controlled comparison. Analyze both successes and failures.

- Deliverables: 15-minute presentation (12 min + 3 min Q&A) and report

- Project grading: 50% presentation + 50% report. Grading rewards sound experiments and clear analysis; outperforming the baseline is not required.

- Tentative schedule:

- P3 handbook release on 10/19
 - Form a group by 11/2
 - Presentation on 12/7; Report and slides due on 12/20

评分 — 细节

- 迟交政策

- 作业与项目结果在截止日期当天满分。除非提前获准延期，之后 48 小时最多 75%，再之后最多 50%。需要延期请尽早提出。及时申请更容易获批。所有作业都必须提交。
- 随堂测验不补考

Grading - Details

- Late Policy

- Homework assignments and course project results are worth full credit on the due date. Unless granted an extension in advance, it is worth at most 75% credit for the next 48 hours, at most 50% credit after that. If you need an extension, please ask for it as soon as the need for it is known. Extensions that are requested promptly can be granted more liberally. You must turn in all assignments.
- **There will be no make-up quizzes**

评分 — 细节

- 使用 LLM/VLM (ChatGPT、Gemini 等)

- 鼓励用 ChatGPT 做作业题 (因为未来大家很可能都用它写代码)。但须在作业 README 中声明使用了 ChatGPT/LLM, 并附上所用提示词
- 随堂测验严禁使用 LLM!!!

Grading - Details

- Use of LLM/VLMs (ChatGPT, Gemini, etc.)

- We encourage you to use ChatGPT to solve the homework assignment problems (because likely in the future everyone will use it for coding). However, you will need to declare using ChatGPT/LLMs in the homework README, as well as the prompts you use
- **You are not allowed to use LLM for in-class test!!!**

更多课程资源

• 教材

- Machine Learning: A Probabilistic Perspective - Kevin P. Murphy
- Multiple View Geometry in Computer Vision - Richard Hartley and Andrew Zisserman
- Introduction to Autonomous Mobile Robots - Illah Reza Nourbakhsh and Roland Siegwart
- 中文教材: 具身智能原理与实践 (我的教授朋友所著)

• 研讨会与竞赛

- Precognition @CVPR [link]
- Workshop on Autonomous Driving @CVPR [link]
- CARLA Autonomous Driving Challenge @NeurIPS [link]

• 阅读材料

○ 精彩报告

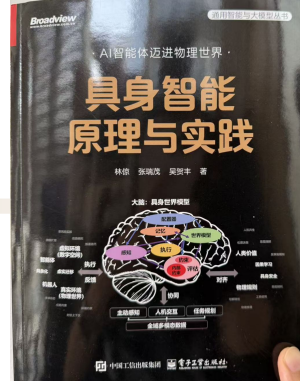
■ A Path Towards Embodied AI - Yann LeCun [link]

■ Robotics Professor Answers Robot Questions From Twitter - Henny Admoni [link]

○ 其他公开课

- Stanford AA 274a
- CMU 16-311, 11-851
- Washington CSE 478
- Texas CS 391R

• 更多在线教程 / 具身智能重要知识 (梁俊伟老师)



More Course Resources

• Textbooks

- Machine Learning: A Probabilistic Perspective - Kevin P. Murphy
- Multiple View Geometry in Computer Vision - Richard Hartley and Andrew Zisserman
- Introduction to Autonomous Mobile Robots - Illah Reza Nourbakhsh and Roland Siegwart
- Chinese textbook: 具身智能原理与实践 by my professor friends

• Workshops and Competitions

- Precognition @CVPR [link]
- Workshop on Autonomous Driving @CVPR [link]
- CARLA Autonomous Driving Challenge @NeurIPS [link]

• Reading Materials

○ Nice talks

■ A Path Towards Embodied AI - Yann LeCun [link]

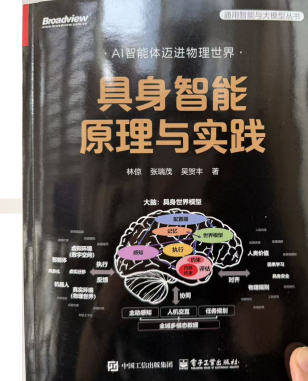
■ Robotics Professor Answers Robot Questions From Twitter - Henny Admoni [link]

○ Other Open Courses

- Stanford AA 274a
- CMU 16-311, 11-851
- Washington CSE 478
- Texas CS 391R

• More online tutorial videos/Important knowledge about Embodied AI by Prof. Junwei Liang

- 【Tencent Docs】Embodied AI学习指引 for UG
- <https://docs.qq.com/sheet/DREN4UUpyaVBvbGdL?tab=qgffmc>



你的待办

- 查看课程网页、Canvas

○ <https://hkust-aiaa4220.github.io/fall2026/index.html>

- 复习线性代数、微积分与概率
- 复习 Python、NumPy、Ubuntu
- 试用三维仿真器

○ <https://carla.org/>

○ <https://behavior.stanford.edu/>

○ <https://developer.nvidia.com/isaac-sim>

- 开始挑选要读的论文

○ 具身智能论文列表: [link](#)

- 开始找课程项目队友!

Your TODOs

- Check course webpage, Canvas

- <https://hkust-aiaa4220.github.io/fall2026/index.html>

- Review linear algebra, calculus and probability

- Review python coding, numpy, Ubuntu system

- Checkout 3D simulators and play around

- <https://carla.org/>

- <https://behavior.stanford.edu/>

- <https://developer.nvidia.com/isaac-sim>

- Start checking out the paper to read

- Embodied AI paper list: [link](#)

- Start finding group members to pair with others in course projects!



感谢你决定与我们一起踏上具身智能之旅
希望你能学到东西，并取得好成绩！
有问题吗？



Thank you for deciding to take this journey of Embodied AI with us
Hope you can learn something and get a good grade!
Any questions?